ERROR PROPAGATION UNDER RECURSIVE ORTHOGONAL PROJECTION OF FINITE MULTILINEAR COMPOSITION TREES

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Abstract. Let T be a finite rooted ordered tree whose internal vertices carry bounded multilinear maps between finite-dimensional Hilbert spaces, and suppose that the output at each internal vertex is orthogonally projected onto a prescribed subspace of its ambient space. We compare the value obtained by unprojected evaluation with the value obtained by recursive projection. We derive an exact local decomposition of the discrepancy into a closure residual created at the vertex and a sum over all nonempty subsets of the children whose subtree errors interact, and we show that at the root the discrepancy splits orthogonally, with the reduced-coordinate error equal to the projected component exactly. Under uniform bounds M on the multilinear maps and ρ on their failure to preserve the prescribed subspaces, we prove that a tree with $k \geq 1$ internal vertices satisfies $E_T^{\mathrm{amb}} \leq k\rho M^{k-1} L_T$ while $E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} \leq (k-1)\rho M^{k-1} L_T$, where L_T is the product of the leaf norms; the reduction from k to $k-1$ is exactly the removal of the root closure residual by the final projection. We also derive state-resolved and pathwise upper bounds, prove that a closed-form ordering rule minimises the telescoping bound within a declared scalar family, and treat finite signed linear combinations of trees. The coefficient $k-1$ is an upper bound; it is not shown to be attained at fixed $\rho/M > 0$. Over a registry of 9,945 configurations, the constant is determined in 309 cases and vanishes by the theorem in 30 further cases, while in 1,794 cases (18.0 %) no positive lower bound was obtained at all. Determining the exact constant, already for two and three internal vertices, remains open.

1. Introduction

Finite compositions of multilinear maps arise in nonassociative algebra, in the syntax of coloured operads, in hierarchical tensor computations, and in projection-based model reduction. In all of these settings a recurring computational step is to project the output of every intermediate operation back onto a prescribed subspace, either because the subspace carries the structure one wishes to preserve, or because storing the full ambient object is too expensive.

Projection is exact only when the prescribed subspaces are invariant under the maps. When they are not, each internal vertex creates a residual lying in the orthogonal complement of its subspace, and these residuals then propagate through the remaining composition. The question studied here is how those local residuals accumulate.

A crude answer is available immediately. Bounding every intermediate quantity by the product of the operator norms yields an estimate that is valid but uninformative: it neither identifies which vertex is responsible for the error, nor distinguishes the part of the error that survives the final projection from the part that does not, nor exploits the geometry of orthogonality at all. The purpose of this article is to replace that estimate by a decomposition that does all three.

What is proved. The starting point is an exact identity. At each internal vertex, the discrepancy between the two evaluations is the sum of a locally created residual and a term for each nonempty

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The proofs and the numerical results of this article have not been checked by anyone other than the author. No claim of originality is made; see Section 2.

subset of the children whose subtree errors are simultaneously active (Theorem 5.1). No inequality is used, and no expansion is truncated. Applying the vertex projector annihilates the locally created residual exactly and leaves the propagated terms untouched.

From this identity three kinds of estimate follow. The first is a uniform estimate: with M bounding the multilinear maps and ρ bounding their closure residuals,

$$E_T^{\mathrm{amb}} \leq k\rho M^{k-1} L_T, \qquad E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} \leq (k-1)\rho M^{k-1} L_T$$

for every tree with $k \geq 1$ internal vertices (Theorem 9.1). The two coefficients differ by exactly one, because the ambient error may contain a residual created at every internal vertex, whereas the residual created at the root lies in the orthogonal complement of the root subspace and is annihilated by the final projection.

The second is a family of vertexwise estimates that track, separately, the projected and orthogonal components of every subtree error (Section 7). These are finer than the uniform estimate because a residual created orthogonally at one vertex can acquire a projected component at its parent, and the two situations carry different constants.

The third attributes the bound to individual vertices: unrolling a valid scalar recurrence expresses the estimate as a sum over internal vertices of a locally created source multiplied by the gains along its path to the root (Section 10).

What is not proved. The coefficient $k-1$ is an upper bound. We do not show that it is the best possible coefficient at a fixed positive value of $\eta = \rho/M$, and the numerical evidence assembled here does not settle the question: explicit constructions attain the ambient coefficient in the configurations where it is determined, while the projected constructions we have found remain strictly below $k-1$ except in a small number of configurations. Section 13 reports what is known for two and three internal vertices; in both cases the exact constant is undetermined over most of the parameter range.

Every result is finite-dimensional. Nothing here is a statement about infinite composition, about continuum limits, or about function spaces.

Organisation. Section 2 places the results relative to the literature. Sections 3 and 4 fix the setting. Section 5 proves the exact local decomposition and Section 6 the orthogonal splitting at the root. Sections 7 to 10 develop the state-resolved, order-optimised and pathwise estimates, and Section 9 the uniform estimate. Section 11 treats signed combinations of trees. Sections 12 and 13 report lower constructions and the unresolved low-order cases. Section 14 adds approximation of the maps themselves. Section 15 describes the numerical verification, Section 17 the limitations and Section 18 the open problems. Complete proofs are in Appendix A and reproduction instructions in Appendix B.

2. Relation to previous work

The typed composition of multilinear operations is the setting of coloured operads [15, 12], from which we borrow only the bookkeeping: the trees here are finite and the operations are concrete linear-algebraic objects. Signed combinations of composition trees are the natural home of the nonassociative identities of Filippov [4] and of the Akivis–Sabinin circle of ideas [14].

The question of how a map that is only approximately structure-preserving behaves under composition is classical in operator algebra: Johnson's work on approximately multiplicative maps between Banach algebras [8] asks a related question under global hypotheses, but does not address a fixed finite tree with prescribed subspaces at every vertex.

Forward and backward error analysis has always treated the order of composition and the propagation of local perturbations as central [7]; the estimates of Sections 7–10 are recognisably of that kind. Layerwise Lipschitz estimates for composed operators [2] and sound abstract interpretation for layered computations [5] show, in different settings, that a structure-aware summary can improve on a plain product of norms. We do not claim that a vertexwise recursion is itself new.

Hierarchical tensor formats organise computation along a dimension tree and adapt the approximation to it [6, 1]; canonical polyadic decomposition, factor nonuniqueness and identifiability are a separate matter [9, 10]. The dominant error in those formats is representation or rank truncation. The dominant error here is different in kind: it is a closure defect created at every algebraic vertex, and Section 14 keeps the two apart deliberately. Projection-based structure-preserving model reduction supplies the closest methodological precedent [3].

Interval arithmetic [13] and moment relaxations [11] provide rigorous global tools when their hypotheses and solver statuses are verified. We use directed interval recursion and, in a few low-dimensional cases, exact symbolic elimination. We do not report a semidefinite relaxation as a certificate.

Remark 2.1 (Originality). We make no claim of originality for any statement in this article. A bounded search of the areas above did not locate the projected-error coefficient, the state-resolved recursion or the pathwise attribution in the form given here, but absence from a bounded search is not evidence of novelty. The relationship between these estimates and existing error bounds has not been assessed sufficiently to support such a claim, and the comparison in Table 1 should be read as a record of overlap, not as a verdict.

3. Typed multilinear composition trees

Let $\mathcal{T}$ be a finite set of *types*. To each $\tau \in \mathcal{T}$ we associate a finite-dimensional Hilbert space V_τ over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, a second finite-dimensional Hilbert space W_τ, and an isometry

$$Q_\tau : W_\tau \longrightarrow V_\tau, \qquad Q_\tau^* Q_\tau = I_{W_\tau}.$$

We call V_τ the *ambient* space and W_τ the *reduced* space of type τ, and we write

$$P_\tau = Q_\tau Q_\tau^*,$$

so that P_τ is the orthogonal projector of V_τ onto $\operatorname{ran} Q_\tau$, and $P_\tau^2 = P_\tau = P_\tau^*$, $\|P_\tau\| = 1$.

Throughout, $\operatorname{ran} P_\tau$ is a fixed linear subspace of V_τ. It is not a tangent space, and we use the words *projected* and *orthogonal* for the two components of a vector relative to it.

Definition 3.1 (Composition tree)**.** A *typed composition tree* T is a finite rooted ordered tree together with: an output type $\tau(v) \in \mathcal{T}$ and an arity $a_v \geq 1$ for each internal vertex v; input types $\tau(v,1), \dots, \tau(v, a_v)$; a multilinear map

$$\mu_v : \prod_{j=1}^{a_v} V_{\tau(v,j)} \longrightarrow V_{\tau(v)};$$

and a type $\tau(\ell) \in \mathcal{T}$ for each leaf ℓ. The jth child of v must have output type $\tau(v,j)$. Trees violating this condition are not admitted: they have no numerical semantics.

We write $\operatorname{Int} T$ for the set of internal vertices, $k(T) = |\operatorname{Int} T|$, and $a_{\max} = \max_v a_v$. The multilinear operator norm is

$$\|\mu\|_{\mathrm{op}} = \sup\{\|\mu(x_1, \dots, x_a)\| : \|x_1\| = \cdots = \|x_a\| = 1\}.$$

Definition 3.2 (Closure-residual map)**.** For an internal vertex v with children $c_1, \dots, c_a$, the *closure-residual map* is the multilinear map

$$r_v = (I - P_{\tau(v)})\, \mu_v\big(P_{\tau(c_1)} \cdot, \dots, P_{\tau(c_a)} \cdot\big).$$

It vanishes identically if and only if $\mu_v(\operatorname{ran} P_{\tau(c_1)}, \dots, \operatorname{ran} P_{\tau(c_a)}) \subseteq \operatorname{ran} P_{\tau(v)}$, that is, if and only if the prescribed subspaces are invariant under μ_v.

We abbreviate $P_v = P_{\tau(v)}$ when no confusion arises.

3.1. **Standing hypotheses.** Every statement in this article is made under the following hypotheses, which are not repeated in the individual statements.

(H1) The scalar field is $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, fixed throughout. Nothing below distinguishes the two cases.
(H2) For each $\tau \in \mathcal{T}$, V_τ and W_τ are finite-dimensional Hilbert spaces over $\mathbb{K}$, with $1 \le \dim W_\tau \le \dim V_\tau$.
(H3) $Q_\tau : W_\tau \to V_\tau$ is an isometry and $P_\tau = Q_\tau Q_\tau^*$ is the associated *orthogonal* projector. Obliqueness is not permitted; see Example 6.2.
(H4) T is a finite rooted ordered tree. Every internal vertex has finite arity $a_v \ge 1$, and $\mathcal{T}$ is finite.
(H5) Every μ_v is multilinear and bounded, with $\|\mu_v\|_{\mathrm{op}} < \infty$. In finite dimension this is automatic and is stated for emphasis.
(H6) The tree is type-compatible in the sense of Definition 3.1. Type-incompatible trees are not admitted and receive no numerical semantics.
(H7) Leaf data z_ℓ lie in the *reduced* spaces $W_{\tau(\ell)}$, so that $F_\ell = R_\ell = Q_{\tau(\ell)} z_\ell \in \operatorname{ran} P_{\tau(\ell)}$.

The three constants have the following exact meaning.

M	any real number with $\|\mu_v\|_{\mathrm{op}} \le M$ for *every* internal vertex v; necessarily $M \ge 0$
ρ	any real number with $\|r_v\|_{\mathrm{op}} \le \rho$ for *every* internal vertex v, r_v being the closure-residual map of Definition 3.2; necessarily $\rho \ge 0$
L_T	$\prod_\ell \|z_\ell\|$, the product over the leaves of the norms of the reduced leaf data

Neither M nor ρ is required to be attained. Since $\|I - P_v\| \le 1$ and $\|P_{\tau(c_i)}\| \le 1$, one has $\|r_v\|_{\mathrm{op}} \le \|\mu_v\|_{\mathrm{op}}$ at every vertex, so ρ may always be chosen with $\rho \le M$; the ratio $\eta = \rho/M$ of Section 13 is understood with that choice and lies in $[0,1]$ when $M > 0$.

3.2. **Degenerate cases.** These are stated once and used without comment.

$k = 0$	T is a single leaf. Then $F = R$ and all four errors vanish. The expression M^{k-1} is not defined at $k = 0$ when $M = 0$, so Theorem 9.1 is stated for $k \ge 1$ and this case is handled separately in Appendix A
$k = 1$	The root is the only internal vertex and every child is a leaf. Then $D_\varrho = r_\varrho(R_1, \dots, R_a)$ lies entirely in $(\operatorname{ran} P)^\perp$, so $E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} = 0$ exactly, consistently with the coefficient $k - 1 = 0$, while $E_T^{\mathrm{amb}} = E_T^\perp$ may be nonzero
$M = 0$	Every μ_v is the zero map, hence $\rho = 0$ and every evaluation and every error vanishes. All estimates read $0 \le 0$ for $k \ge 1$, with the convention $0^0 = 1$ at $k = 1$
$\rho = 0$	Every prescribed subspace is invariant under its vertex map. Then $D_v = 0$ at every vertex by Theorem 5.1 and induction, projection is exact, and all four errors vanish. The estimates read $0 \le 0$
$a_v = 1$	Permitted: a unary vertex carries a bounded linear map. All statements hold, and the interaction terms of Theorem 5.1 are absent because $[a_v]$ has only one nonempty subset
$L_T = 0$	Some leaf datum is zero; multilinearity makes every evaluation zero and the estimates read $0 \le 0$

4. Unprojected and recursively projected evaluation

Fix reduced leaf data $z_\ell \in W_{\tau(\ell)}$ and set

$$L_T = \prod_{\ell \in \mathrm{Leaves}(T)} \|z_\ell\|.$$

Definition 4.1 (The two evaluations)**.** The *unprojected* (or ambient) evaluation is

$$F_\ell = Q_{\tau(\ell)} z_\ell, \qquad F_v = \mu_v(F_{c_1}, \dots, F_{c_{a_v}}),$$

and the *recursively projected* evaluation is

$$R_\ell = Q_{\tau(\ell)} z_\ell, \qquad R_v = P_{\tau(v)}\, \mu_v(R_{c_1}, \dots, R_{c_{a_v}}).$$

The *local discrepancy* at v is $D_v = F_v - R_v$, with components

$$D_v^{\parallel} = P_v D_v, \qquad D_v^{\perp} = (I - P_v) D_v.$$

By construction $R_v \in \operatorname{ran} P_v$ for every internal v, and $R_\ell \in \operatorname{ran} P_{\tau(\ell)}$ at every leaf.

Write ϱ for the root of T and abbreviate $F = F_\varrho$, $R = R_\varrho$, $P = P_{\tau(\varrho)}$, $Q = Q_{\tau(\varrho)}$. The four quantities of interest are

$$E_T^{\text{amb}} = \|F - R\|, \quad E_T^{\text{proj}} = \|PF - R\|, \quad E_T^{\perp} = \|(I - P)F\|, \quad E_T^{\text{red}} = \|Q^* F - Q^* R\|.$$

The first is the discrepancy in the ambient space; the second is what remains of it after the final projection; the third is what the final projection discards; the fourth is the discrepancy as measured in the reduced coordinates, which is what a computation carried out entirely in the reduced spaces would report.

5. Exact local decomposition

Theorem 5.1 (Exact local decomposition)**.** *Let v be an internal vertex of arity a with children $c_1, \ldots, c_a$, and write $F_i = F_{c_i}$, $R_i = R_{c_i}$, $D_i = F_i - R_i$. For $S \subseteq [a]$ put*

$$y_i^S = \begin{cases} D_i, & i \in S, \\ R_i, & i \notin S. \end{cases}$$

Then

$$D_v = r_v(R_1, \ldots, R_a) + \sum_{\varnothing \neq S \subseteq [a]} \mu_v(y_1^S, \ldots, y_a^S).$$

The identity is exact: no term is discarded and no inequality is used. The term indexed by S records precisely which children carry an error. Singletons are single-branch propagation; subsets of size at least two are genuine interactions between the errors of different branches.

Two consequences are used repeatedly. Since $P_v r_v = 0$,

$$D_v^{\parallel} = \sum_{\varnothing \neq S \subseteq [a]} P_v \mu_v(y_1^S, \ldots, y_a^S) :$$

the residual created at a vertex never appears in that vertex's own projected error. Conversely, μ_v need not map the orthogonal complement into itself, so a child's orthogonal error can contribute to its parent's projected error,

$$P_v \mu_v(\ldots, D_i^{\perp}, \ldots) \neq 0 \quad \text{in general.}$$

Both effects are visible in the constants of Section 9.

6. Orthogonal splitting at the root

Proposition 6.1 (Root geometry)**.** *For every typed composition tree,*

$$\left(E_T^{\text{amb}}\right)^2 = \left(E_T^{\text{proj}}\right)^2 + \left(E_T^{\perp}\right)^2, \qquad E_T^{\text{red}} = E_T^{\text{proj}}.$$

Both are identities. The first is the Pythagorean theorem applied to $F - R = (PF - R) + (I - P)F$, whose two summands lie in $\operatorname{ran} P$ and $(\operatorname{ran} P)^{\perp}$ respectively; the second says that a computation performed in the reduced coordinates observes exactly the projected component of the ambient discrepancy, no more and no less.

Orthogonality of the projectors is used, and cannot be dropped.

Example 6.2 (Failure for an oblique projector)**.** Let $P = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ on $\mathbb{R}^2$. Then $P^2 = P$, so P is a projector, but $P^* \neq P$. For $D = (0, 1)^{\top}$ one has $PD = (1, 0)^{\top}$ and $(I - P)D = (-1, 1)^{\top}$, whence $\langle PD, (I - P)D \rangle = -1 \neq 0$: the two components are not orthogonal and Proposition 6.1 fails.

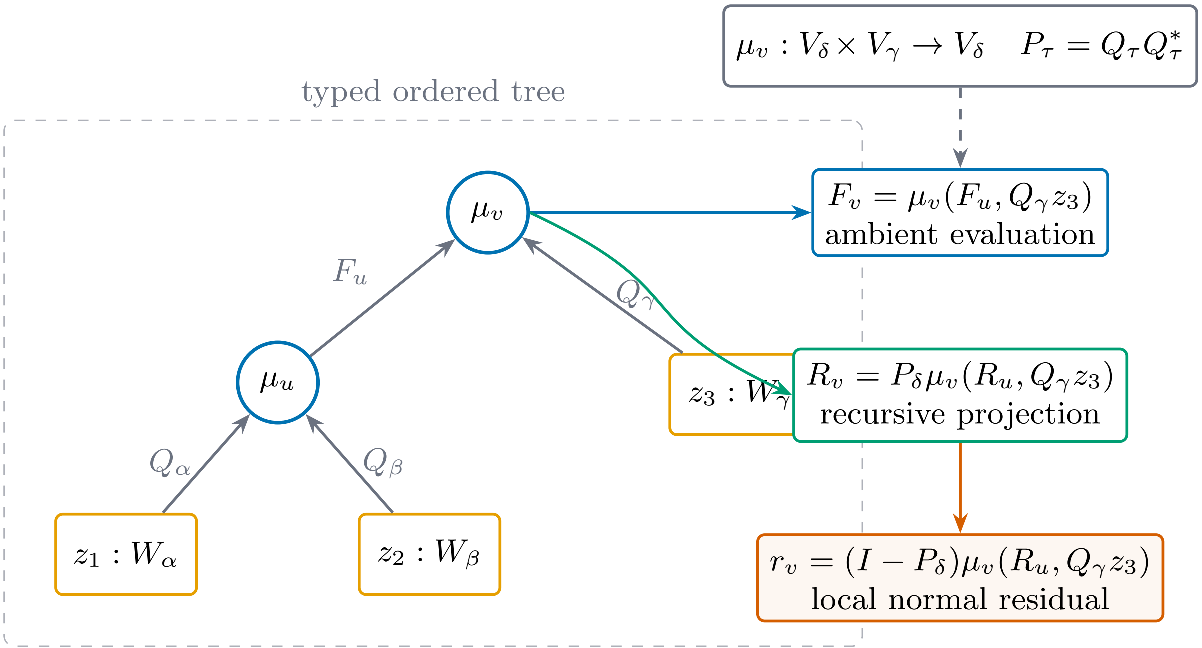

Figure 1. Semantics of a typed ordered composition tree. Types determine the ambient spaces, the isometric embeddings, the orthogonal projectors and the admissible multilinear maps; unprojected evaluation, recursive projection and the local closure residual are three distinct operations. This is a definitional diagram: it carries no sampling uncertainty and supports no statement about optimal constants.

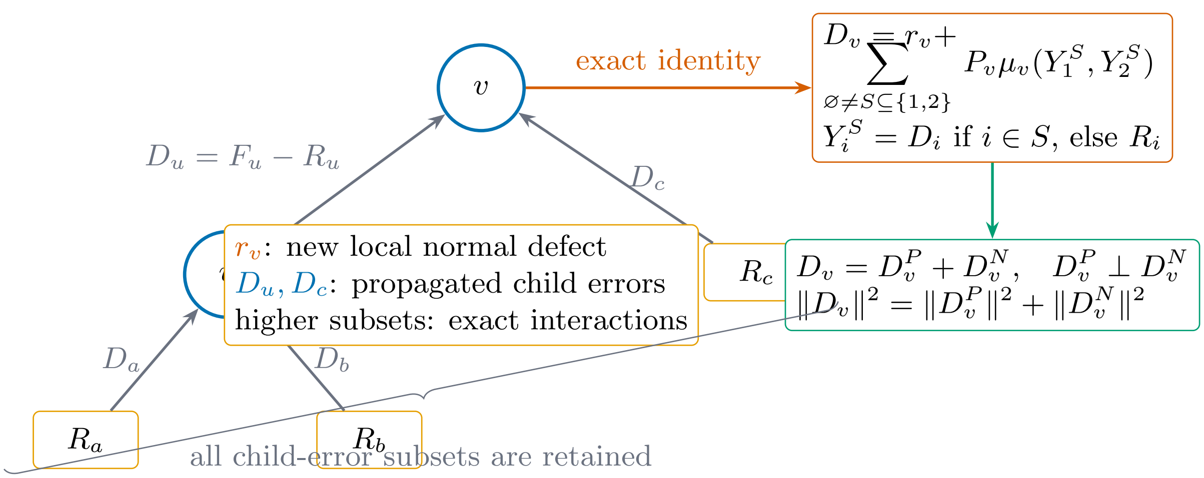

Figure 2. Exact local error decomposition. The local closure residual and every nonempty subset of propagated child errors appear in the multilinear identity, followed by the orthogonal splitting into projected and orthogonal components. This is a diagram of a proof; it is an identity, not a bound, and it does not determine optimal constants.

7. State-resolved estimates

Splitting each child discrepancy as $D_i = D_i^{\|} + D_i^{\perp}$ and substituting into Theorem 5.1 expresses the value at v as a sum over states

$$\mu_v(F_1, \dots, F_a) = \sum_{\sigma \in \{R, \|, \perp\}^a} \mu_v(Z_1^{\sigma}, \dots, Z_a^{\sigma}), \qquad Z_i^{\sigma} = \begin{cases} R_i, & \sigma_i = R, \\ D_i^{\|}, & \sigma_i = \|, \\ D_i^{\perp}, & \sigma_i = \perp . \end{cases}$$

The state vector $(R, \dots, R)$ contains the reduced value together with the locally created residual; every other state vector contributes propagation or interaction.

Let $\beta_v^{\chi}(\sigma)$ denote the operator norm of μ_v restricted to the indicated components of its arguments and projected onto the component χ of its output, for $\chi \in \{\|, \perp\}$. The R and $\|$ states are carried by the same projector but obey different magnitude bounds, which is why they are tracked separately.

Theorem 7.1 (Soundness of the state-resolved recursion)**.** *Assume that for every internal vertex v and every state vector $\sigma \in \{R, \|, \perp\}^{a_v}$ an upper bound for $\beta_v^{\chi}(\sigma)$ is available. Then the recursion that stores, at each vertex, upper bounds for $\|F_v\|$, $\|R_v\|$, $\|D_v\|$, $\|D_v^{\|}\|$ and $\|D_v^{\perp}\|$, and combines them by*

$$b_v^{\chi} \le \lambda_v^{\chi} + \sum_{\sigma \ne (R, \dots, R)} \beta_v^{\chi}(\sigma) \prod_i b_{c_i}^{\sigma_i}, \qquad \lambda_v^{\|} = 0, \quad \lambda_v^{\perp} \le \rho_v \prod_i b_{c_i}^{R},$$

returns valid upper bounds at the root. It performs $O(3^{a_v})$ combination steps at each vertex v, hence $O(|T|\, 3^{a_{\max}})$ in total.

Remark 7.2. The count in Theorem 7.1 is a count of *combination steps in the recursion.* It excludes the cost of obtaining the bounds $\beta_v^{\chi}(\sigma)$, which dominates in practice and which we do not analyse; see Section 17.

8. Optimal ordering for multilinear telescoping

Comparing $\mu(F_1, \dots, F_a)$ with $\mu(R_1, \dots, R_a)$ by exchanging one argument at a time produces a bound that depends on the order of exchange. Let $e_i \ge 0$ bound the error in slot i, let $r_i, f_i \ge 0$ bound $\|R_i\|$ and $\|F_i\|$, let $G_i \ge 0$ be a slot gain, and put $w_i = G_i e_i$. For a permutation π of $[a]$ the resulting bound is

$$C(\pi) = \sum_{t=1}^{a} w_{\pi_t} \prod_{s<t} r_{\pi_s} \prod_{s>t} f_{\pi_s}.$$

Theorem 8.1 (Optimal order)**.** *Put $d_i = f_i - r_i$. Order the slots by placing those with $d_i > 0$ first, then those with $d_i = 0$, then those with $d_i < 0$, and sorting each of the two nonzero sign classes by nondecreasing w_i/d_i. The resulting permutation minimises C over all permutations. Slots with $w_i = d_i = 0$ may be placed anywhere.*

The proof is an adjacent-exchange argument: placing i before j is no worse precisely when $w_i d_j \le w_j d_i$. What makes the argument yield a *global* minimum, rather than only a local one, is that this comparison is induced by the lexicographic key (sign class of d_i, w_i/d_i), with sign classes ordered positive, zero, negative, and is therefore a total preorder; the details are in Appendix A.

Remark 8.2. Theorem 8.1 is optimal within the family of scalar telescoping bounds defined by C. It says nothing about bounding strategies outside that family.

9. Uniform estimates

Theorem 9.1 (Uniform coefficients). *Suppose that $\|\mu_v\|_{\mathrm{op}} \le M$ and $\|r_v\|_{\mathrm{op}} \le \rho$ at every internal vertex. Then for every typed composition tree with $k = k(T) \ge 1$ internal vertices,*

$$E_T^{\mathrm{amb}} \le k\rho M^{k-1} L_T, \qquad E_T^{\perp} \le k\rho M^{k-1} L_T,$$

and

$$E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} \le (k-1)\rho M^{k-1} L_T.$$

The hypotheses are uniform over the vertices, but nothing else is assumed to be uniform: types, arities, dimensions, ranks and the maps themselves may all differ from vertex to vertex.

Remark 9.2 (The case of a leaf). The statement is made for $k \ge 1$. A leaf has $k = 0$ and error 0, and the expression M^{k-1} is not defined at $k = 0$ when $M = 0$; the induction in Appendix A treats the leaf case separately, where the assertion reads $0 \le 0$.

Sketch of proof. Induction on the tree. At an internal vertex, multilinear telescoping over the children together with the magnitude estimate $\|F_v\|, \|R_v\| \le M^{k_v} L_v$ of Lemma A.1 bounds the propagated contribution by $\left(\sum_i k_{c_i}\right)\rho M^{k_v-1} L_v = (k_v - 1)\rho M^{k_v-1} L_v$. The locally created residual adds at most $\rho \prod_i \|R_i\| \le \rho M^{k_v-1} L_v$, which gives the ambient estimate. For the projected estimate, $R \in \operatorname{ran} P$ gives $PF - R = PD_\varrho$, and $Pr_\varrho = 0$ removes the root residual, so only the propagated contribution survives. Proposition 6.1 identifies the reduced error with the projected one. The orthogonal estimate follows from $(I-P)F = (I-P)D_\varrho$ and $\|I - P\| = 1$. Complete details are in Appendix A. □

Remark 9.3 (Where the difference of one comes from). The reduction from k to $k-1$ is not a refinement of the estimate: it is the exact removal of one source. The residual created at the root lies in $(\operatorname{ran} P)^\perp$ and is annihilated by the final projection. In the extreme case $k = 1$, every child is a leaf, so $D_\varrho = r_\varrho(R_1, \ldots, R_a)$ lies entirely in the orthogonal complement, and

$$E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} = 0$$

exactly, while E_T^{amb} need not vanish. The estimate is consistent: $(k-1)\rho M^{k-1} L_T = 0$.

Remark 9.4 (Upper bound, not optimal constant). Theorem 9.1 is a chain of triangle inequalities. It establishes upper bounds and nothing more. In particular it does not show that $k - 1$ is the smallest coefficient valid at a fixed value of $\eta = \rho/M > 0$, nor for a fixed dimension, projector rank, topology, or class of repeated maps. Sections 12 and 13 report what is and is not known about this.

10. Pathwise estimates

Fix a valid ordering at every vertex, so that the quantities B_v produced by the recursion of Section 7 satisfy the affine scalar inequality

$$B_v \le \lambda_v + \sum_j h_{v,j} B_{c_j}, \qquad \lambda_v = \rho_v \prod_i B_{c_i}^R, \qquad B_\ell = 0, \tag{1}$$

where $h_{v,j} \ge 0$ is the gain for transporting an error from slot j through v.

The right-hand side of (1) is an inequality, so B_ϱ itself has no closed form. We therefore introduce the quantity that does.

Definition 10.1 (Pathwise majorant). Let $\widehat{B}_v$ be defined by the *equality*

$$\widehat{B}_\ell = 0, \qquad \widehat{B}_v = \lambda_v + \sum_j h_{v,j} \widehat{B}_{c_j} \quad (v \in \operatorname{Int} T).$$

Corollary 10.2 (Pathwise attribution)**.** *For every vertex v one has $B_v \le \widehat{B}_v$, and at the root*

$$\widehat{B}_\varrho = \sum_{v \in \operatorname{Int} T} \lambda_v \prod_{e \in \operatorname{path}(v,\varrho)} h_e,$$

the product being over the edges from v to the root, each contributing the gain of the slot through which it passes, and the product over the empty path being 1. Consequently

$$B_\varrho \le \sum_{v \in \operatorname{Int} T} \lambda_v \prod_{e \in \operatorname{path}(v,\varrho)} h_e.$$

For the projected error the root source is omitted, λ_ϱ being replaced by 0, and the final edge carries a projected-output gain.

Remark 10.3 (What the pathwise formula is, and is not). Two distinctions matter here, and the earlier formulation blurred both.

First, the displayed formula is an *identity for* $\widehat{B}_\varrho$, not for B_ϱ. The recursion of Section 7 produces upper bounds, so (1) is an inequality; unrolling an inequality does not yield an equation. Definition 10.1 supplies the majorant for which the unrolling is exact, and the two statements of Corollary 10.2 are then separate: a comparison $B_v \le \widehat{B}_v$, proved by induction, and a closed form for $\widehat{B}_\varrho$.

Second, even as a statement about $\widehat{B}$, the formula is not an expansion of the multilinear cross terms of Theorem 5.1. The interaction terms indexed by subsets of size at least two were absorbed into the gains $h_{v,j}$ when (1) was formed. Each summand therefore attributes a share of the *bound* to a vertex, not a share of the error.

11. Signed combinations of trees

Let $\mathcal{F} = \sum_{\alpha=1}^{m} c_\alpha T_\alpha$ be a finite linear combination of composition trees with compatible root types.

Corollary 11.1 (Triangle bound)**.** $E^{\mathrm{proj}}_{\mathcal{F}} \le \sum_\alpha |c_\alpha| B^{\mathrm{proj}}_{T_\alpha}$. *In particular, the five-input ternary associator $\mu \circ_1 \mu - \mu \circ_3 \mu$, a difference of two trees with two internal vertices each, satisfies $E^{\mathrm{proj}} \le 2\rho M L$, whereas the corresponding ambient bound is $4\rho M L$.*

Combining syntactically identical trees before applying the triangle inequality is an exact simplification, and it is the only unconditional cancellation available here. Any further cancellation — from shared subtrees, opposite signs, symmetry or antisymmetry — is not captured by Corollary 11.1, and quantifying it is open (Problem 18.3).

11.1. **Numerical lower bounds for five named combinations.**

Table 7 reports the largest values found by numerical search for five combinations. The search used 4,000 random tensor trials followed by 200 derivative-free local refinement steps for each combination, with the multilinear operator norm normalised numerically to approximately 1.

Leaf inputs were drawn independently. This differs from the convention used in the earlier evaluation, which supplied the same unit input at every leaf; identical inputs can collapse permutation-antisymmetric terms for reasons unrelated to the map, and the change was made for that reason.

For the Jacobi combination the search did not improve on the triangle bound in any trial. This is informative in the negative direction: it indicates that reasoning which accounts for cancellation does not help for that combination in the tested regime. It does not establish that the constant equals 3.

11.2. **A quantity that vanishes numerically.** For the declared six-term generalised Jacobi expression, the ambient, projected and orthogonal errors all evaluated between 10^{-16} and 10^{-21} in every one of the 4,000 trials, in five independently reseeded checks, and under both leaf-input conventions.

Resolved, follow-up session. The apparent vanishing above is *not* a general identity. Two independent exact symbolic methods (a fully generic symbolic tensor law with free symbolic leaf vectors, verified by two structurally unrelated implementations) prove the expression is nonzero for generic, non-collinear leaves — an explicit exact-rational counterexample (η-free, dimension 2, no floating point) gives the value $(97/3, 97/3) \neq 0$. The numerical vanishing above is fully explained: every one of the 4,000 trials used the pipeline's rank-one reduced-input convention, which lifts every leaf to a scalar multiple of a single fixed ambient vector. We prove that this specific six-term construction vanishes identically for *any* law whenever all leaves are collinear in this sense — a genuine but considerably narrower fact than a formal identity, which fully accounts for the observed numerics without supporting the general claim. Full detail, both symbolic implementations, the exact counterexample, and mutation tests are in `research/math_closure/gji/`.

12. Lower constructions

An attained value is a rigorous lower bound only once admissibility has been verified. The principal explicit family acts on a two-dimensional plane spanned by one direction in $\operatorname{ran} P$ and one in its orthogonal complement. For $0 \leq \eta \leq 1$ put

$$G_\eta = \begin{bmatrix} \sqrt{1-\eta^2} & -\eta \\ \eta & \sqrt{1-\eta^2} \end{bmatrix}.$$

The first argument is rotated by G_η while the remaining arguments act through their first coordinate. The multilinear operator norm of the resulting map is one and the norm of its closure-residual map is exactly η. Embedding the plane isometrically into a larger ambient space preserves the value.

Directed interval recursion in 160-bit arithmetic encloses the normalised tree error of this family. In a small number of low-dimensional cases an independent exact symbolic routine enumerates the endpoints and stationary points on $[-1, 1]$ and retains an exact nonnegativity witness.

13. Low-order cases and unresolved gaps

Define, for a fixed tree T and a fixed admissible class of data,

$$C_T^{\mathrm{proj}}(\eta) = \sup \frac{E_T^{\mathrm{proj}}}{\rho M^{k-1} L_T}, \qquad \eta = \rho/M,$$

the supremum being over admissible data. Theorem 9.1 states $C_T^{\mathrm{proj}}(\eta) \leq k-1$. Whether equality holds is open.

Table 10 classifies the 9,945 configurations of the enclosure registry into four mutually exclusive outcomes. Over the whole registry:

- in 309 configurations the interval-enclosed lower construction meets the proved upper bound at a positive value, so the constant is determined;
- in 30 configurations the proved upper bound is itself zero. These are the single-internal-vertex cases of Remark 9.3, where the projected error vanishes as an immediate consequence of the theorem. They are not the outcome of an optimisation;
- in 7,812 configurations a positive lower bound was obtained but does not meet the upper bound;
- in 1,794 configurations (18.0 % of the registry) *no positive lower bound was obtained at all.* For these the admissible range spans the whole interval from zero to the proved upper bound, and the largest such upper bound is 32.

13.1. **Two internal vertices.** Re-examining the enclosure registry directly, at $k = 2$ the projected-error bound is attained in 6 of 75 uniform gated-rotation configurations. Elsewhere the gaps range from 3.75×10^{-7}, in the limit $\eta \downarrow 0$, up to the whole interval. No ambient or orthogonal configuration at $k = 2$ attains its bound.

Accordingly: *for trees with two internal vertices the exact constant is known only on part of the admissible parameter range; on the remainder the available rigorous lower and upper bounds do not coincide.*

Exact closed form, follow-up session. For the homogeneous chain ($k = 2$, gated-planar-rotation law, unit leaf inputs), the projected error is *exactly* η^2 for every $\eta \in [0, 1]$, independent of the ambient dimension $n \geq 2$ and reduced rank $1 \leq r \leq n - 1$ — proved by exact symbolic substitution through the reference evaluator (not floating point), cross-checked at four distinct (n, r) pairs. Consequently this construction saturates the universal bound if and only if $\eta = 1$, with a linear approach for $\eta < 1$: this is the exact mechanism behind the six exact rows above, all of which occur at $\eta = 1$. Full proof in `research/math_closure/k2/classification_theorem.tex`.

13.2. **Three internal vertices.** At $k = 3$ no configuration and no error type attains its bound, and the smallest projected relative gap observed is 35 %.

Accordingly: *for three internal vertices the available bounds depend on tree topology and leave nonzero gaps throughout.*

Exact closed forms and a corrected optimum, follow-up session. For the same gated-rotation law, the chain topology (three nodes in series) has projected error exactly $3\eta^2\sqrt{1 - \eta^2}$, and the branching topology ($u_1, u_2 \to$ root) exactly $\eta^2\sqrt{1 - \eta^2}$ — both proved by exact symbolic substitution. Both are maximised at $\eta^\star = 1/\sqrt{2}$, giving *exact* best-achievable ratios of 3/4 (chain) and 1/4 (branching) to the universal bound: neither topology saturates the bound at any η, and chain is exactly three times as efficient as branching at the shared optimum. This corrects the 35 % figure above: that number came from sampling only $\eta \in \{0.001, 0.01, 0.1, 0.5, 1\}$, which never tested $\eta^\star \approx 0.707$; the true continuous minimum relative gap is 1/4, not 0.35. Full proofs in `research/math_closure/k3/`.

14. Approximation of the maps

Suppose each μ_v is replaced by an approximation $\widehat{\mu}_v$ with $\|\mu_v - \widehat{\mu}_v\|_{\mathrm{op}} \leq \delta$, so that $\|\widehat{\mu}_v\|_{\mathrm{op}} \leq M + \delta$. This models a low-rank or quantised representation of the maps, and it introduces an error of a different kind from the closure residual. Write F_μ, R_μ for the two evaluations of the exact tree and $R_{\widehat{\mu}}$ for the recursively projected evaluation of the approximate tree.

Proposition 14.1 (Representation and projection)**.** *Under the standing hypotheses of Section 3 for the exact maps $\{\mu_v\}$, and with $\|\mu_v - \widehat{\mu}_v\|_{\mathrm{op}} \leq \delta$ at every internal vertex, a tree with $k \geq 1$ internal vertices satisfies*

$$\|F_\mu - R_{\widehat{\mu}}\| \leq \underbrace{k\rho M^{k-1} L_T}_{closure} + \underbrace{k\delta(M + \delta)^{k-1} L_T}_{representation},$$

and

$$\|PF_\mu - R_{\widehat{\mu}}\| \leq \underbrace{(k-1)\rho M^{k-1} L_T}_{closure} + \underbrace{k\delta(M + \delta)^{k-1} L_T}_{representation}.$$

Remark 14.2 (No closure hypothesis is imposed on the approximate maps)*.* The bound is obtained by inserting R_μ, not $F_{\widehat{\mu}}$:

$$\|F_\mu - R_{\widehat{\mu}}\| \leq \|F_\mu - R_\mu\| + \|R_\mu - R_{\widehat{\mu}}\|.$$

The first term is Theorem 9.1 applied to the *exact* tree, and the second compares two recursively projected evaluations, which differ only in the maps. The alternative route — inserting the unprojected approximate tree $F_{\widehat{\mu}}$ and applying Theorem 9.1 to it — would require a bound on $\|(I - P_v)\widehat{\mu}_v(P\,\cdot, \ldots, P\,\cdot)\|_{\mathrm{op}}$, that is, a closure hypothesis on the *approximate* maps. No such hypothesis is available: δ bounds the distance between μ_v and $\widehat{\mu}_v$ and says nothing about how $\widehat{\mu}_v$ interacts with the prescribed subspaces. The route taken here uses only hypotheses that have been declared.

The two displayed terms are an algebraic partition of one valid upper bound. They are not a decomposition of the observed error, and no equality with it is asserted. In particular the reconstruction error of a low-rank representation is not folded into ρ, and no closure property is attributed to the approximation.

15. Numerical verification

The estimates above were evaluated on 15,493 distinct configurations, comprising 81,445 enumerated tree occurrences (80,870 distinct trees) and 1,530 exhaustive subset evaluations. Repeated executions of the same configuration are not counted as distinct configurations. Optimiser seeds and restarts that were specified but not executed are recorded as not executed rather than omitted.

The largest observed difference between the CPU and GPU evaluators was 1.922×10^{-8}; this is a validation threshold on the implementation, not a bound in any theorem. Bound violations were checked on every configuration and none was found; this tests the implementation of the estimates, not their proofs.

16. Negative controls

The hypotheses are operational rather than decorative, and each is tested by a control that fails when it is dropped.

(1) If the projector is oblique, Proposition 6.1 fails (Example 6.2).
(2) A type-invalid edge cannot be repaired by numerical broadcasting. All 48 declared invalid configurations are rejected before evaluation.
(3) The projected constant of a single-vertex tree is exactly zero, which refutes any formula that assigns the locally created residual to the projected error of the vertex that creates it.
(4) Cancellation within a signed combination can make the triangle bound strict; a numerical lower bound never reverses the direction of an upper bound.
(5) A larger ambient space contains the two-dimensional construction isometrically, but that embedding alone gives no evidence that additional coordinates improve the constant.

17. Limitations

(1) The exact constants at fixed $\eta > 0$ are unknown for most projected, typed, repeated-map and signed-combination families.
(2) Interval enclosures certify the value of an explicit construction. They do not certify a supremum unless a matching upper bound closes the gap.
(3) The specified optimiser grid was not executed in full, under a stated computational-cost constraint. The executed portion is reported as such.
(4) Multilinear operator norms are computed exactly only for the declared explicit constructions. For general dense and low-rank maps a Frobenius upper bound is substituted, which inflates M and therefore every constant that depends on it.
(5) The complexity statement in Theorem 7.1 counts recursion steps and excludes the cost of obtaining the block norms it assumes.
(6) All results are finite-dimensional. No statement is made about continuum limits, infinite trees, measurability, or the behaviour of these estimates under discretisation.
(7) Neither the proofs nor the numerical results have been checked by anyone other than the author, and no assessment of originality has been carried out (Remark 2.1).

18. Open problems

Problem 18.1 (Exact projected constant). Determine $C_T^{\mathrm{proj}}(\eta)$ for nontrivial families of topologies. Deciding between $C_T^{\mathrm{proj}}(\eta) = k-1$ for all admissible η; $C_T^{\mathrm{proj}}(\eta) < k-1$ for $\eta > 0$ with $\lim_{\eta\downarrow 0} C_T^{\mathrm{proj}}(\eta) = k-1$; and $C_T^{\mathrm{proj}}(\eta) \le \Gamma_T(\eta) < k-1$ for some smaller function Γ_T, would each be a substantive result. Nothing in this article favours one over the others. The case $k = 2$ should be settled first.

Problem 18.2 (Dimension and rank). Decide whether ambient dimension or projector rank can improve on the isometrically embedded planar constructions of Section 12.

Problem 18.3 (Cancellation). Obtain rigorous constants for the associator, the Filippov identity and the Jacobi combination that account for cancellation between terms, rather than bounding each term separately.

Problem 18.4 (Symbolic status of the six-term expression). Decide, by exact expansion in the free operad, whether the declared six-term expression of Section 11.2 is identically zero.

Problem 18.5 (Comparison of the estimates). Characterise when the state-resolved, pathwise and order-optimised estimates dominate one another. The ratios in Table 4 are medians over configurations and do not establish an ordering.

Problem 18.6 (Operator norms). Replace the Frobenius upper bounds by validated enclosures of the multilinear spectral norm at a useful scale.

19. Conclusion

Recursive orthogonal projection on a finite typed multilinear tree admits an exact local error decomposition. From it follow a sound state-resolved recursion, a closed-form optimal ordering within a declared scalar family, a pathwise attribution of the bound to individual vertices, and the uniform distinction between k ambient sources and $k-1$ projected sources. The last of these is an upper bound whose optimality is open already for two internal vertices, and the numerical evidence assembled here is broad enough to expose the dependence on topology, interaction, precision and representation without resolving it.

Appendix A. Proofs

This appendix contains the complete finite-dimensional arguments. No statement below depends on an optimiser, a conjecture, or a numerical plot.

A.1. **Magnitudes and the orthogonal splitting at the root.** For a subtree rooted at v, let k_v be its number of internal vertices and L_v the product of the reduced leaf norms below it.

Lemma A.1 (Subtree magnitudes). *If $\|\mu_u\|_{\mathrm{op}} \le M$ at every internal vertex u of the subtree, then*

$$\|F_v\| \le M^{k_v} L_v, \qquad \|R_v\| \le M^{k_v} L_v.$$

Proof. At a leaf, $F_v = R_v = Q_{\tau(v)} z_v$ and $Q_{\tau(v)}$ is an isometry, so both norms equal $\|z_v\| = L_v$, and $k_v = 0$. Suppose the claim holds for the children $c_1, \dots, c_a$ of an internal vertex v. Multilinearity and the definition of the operator norm give

$$\|F_v\| \le M \prod_i \|F_{c_i}\| \le M^{1+\sum_i k_{c_i}} \prod_i L_{c_i} = M^{k_v} L_v.$$

The same argument bounds $\|R_v\| = \|P_v \mu_v(R_{c_1}, \dots, R_{c_a})\|$, because an orthogonal projector is a contraction. Induction proves both claims. $\square$

Proof of Proposition 6.1. At the root write $P = QQ^*$, $F = F_\varrho$ and $R = R_\varrho$. If ϱ is internal then $R = P\mu_\varrho(\cdots) \in \operatorname{ran} P$; if T consists of a single leaf then $R = Qz \in \operatorname{ran} P$ as well. In either case

$$F - R = (PF - R) + (I - P)F,$$

where the first summand lies in $\operatorname{ran} P$ and the second in $(\operatorname{ran} P)^\perp$. The Pythagorean theorem gives the first identity. For the second, $Q^*(I - P) = Q^* - Q^*QQ^* = 0$, so

$$Q^*F - Q^*R = Q^*(PF - R),$$

and the restriction of Q^* to $\operatorname{ran} P$ is an isometry onto $W_{\tau(\varrho)}$, because $Q^*Q = I$. Hence $\|Q^*F - Q^*R\| = \|PF - R\|$. □

Both steps use $P^* = P$. Example 6.2 shows that the conclusion fails for an idempotent that is not self-adjoint.

A.2. **The exact local decomposition.**

Proof of Theorem 5.1. Since $F_i = R_i + D_i$, repeated use of multilinearity in each argument gives

$$\mu_v(F_1, \ldots, F_a) = \sum_{S \subseteq [a]} \mu_v(y_1^S, \ldots, y_a^S),$$

a sum of 2^a terms. The term indexed by $S = \varnothing$ is $\mu_v(R_1, \ldots, R_a)$, while $R_v = P_v\mu_v(R_1, \ldots, R_a)$. Subtracting,

$$D_v = F_v - R_v = (I - P_v)\mu_v(R_1, \ldots, R_a) + \sum_{\varnothing \neq S \subseteq [a]} \mu_v(y_1^S, \ldots, y_a^S).$$

It remains to identify the first summand with $r_v(R_1, \ldots, R_a)$. By construction $R_i \in \operatorname{ran} P_{\tau(c_i)}$ for every i, so $P_{\tau(c_i)}R_i = R_i$, and therefore

$$r_v(R_1, \ldots, R_a) = (I - P_v)\mu_v(P_{\tau(c_1)}R_1, \ldots, P_{\tau(c_a)}R_a) = (I - P_v)\mu_v(R_1, \ldots, R_a).$$

No inequality has been used. □

Remark A.2. The inner projectors in Definition 3.2 are not decorative. The hypothesis $\|r_v\|_{\mathrm{op}} \leq \rho$ of Theorem 9.1 bounds the operator $(I - P_v)\mu_v(P\,\cdot, \ldots, P\,\cdot)$, whose operator norm is in general strictly smaller than that of $(I - P_v)\mu_v(\cdot, \ldots, \cdot)$. The step $P_{\tau(c_i)}R_i = R_i$ above is exactly what permits the smaller quantity to be used.

Applying P_v to Theorem 5.1 removes the locally created residual exactly; applying $I - P_v$ retains it. Splitting each D_i orthogonally into $D_i^{\|} + D_i^{\perp}$ produces the three states $\{R, \|, \perp\}$ used in Section 7.

A.3. **Uniform coefficients.**

Proof of Theorem 9.1. We prove, by induction on the tree, that every subtree rooted at v satisfies

$$\|D_v\| \leq k_v\rho M^{k_v - 1}L_v \quad (k_v \geq 1), \qquad \text{and} \qquad \|D_v\| = 0 \quad (k_v = 0). \tag{2}$$

The two cases are stated separately because M^{k-1} is not defined at $k = 0$ when $M = 0$; compare Remark 9.2.

Base case. At a leaf, $F_v = R_v$, so $\|D_v\| = 0$ and $k_v = 0$.

Inductive step. Let v be internal of arity a with children $c_1, \ldots, c_a$, so that $k_v = 1 + \sum_i k_{c_i}$. Telescoping the arguments of μ_v one at a time, in any fixed order, and using multilinearity together with $\|\mu_v\|_{\mathrm{op}} \leq M$,

$$\|\mu_v(F_1, \ldots, F_a) - \mu_v(R_1, \ldots, R_a)\| \leq M \sum_{j=1}^{a} \|F_j - R_j\| \prod_{i<j} \|R_i\| \prod_{i>j} \|F_i\|. \tag{3}$$

Fix j. If $k_{c_j} = 0$ then $\|F_j - R_j\| = 0$ and the jth summand vanishes, which agrees with $k_{c_j}\rho M^{k_v-1}L_v = 0$. If $k_{c_j} \geq 1$, the inductive hypothesis and Lemma A.1 bound the jth summand by

$$M \cdot k_{c_j}\rho M^{k_{c_j}-1}L_{c_j} \cdot \prod_{i\neq j} M^{k_{c_i}}L_{c_i} = k_{c_j}\rho M^{\sum_i k_{c_i}}L_v = k_{c_j}\rho M^{k_v-1}L_v.$$

Summing over j and using $\sum_j k_{c_j} = k_v - 1$, the right-hand side of (3) is at most $(k_v-1)\rho M^{k_v-1}L_v$.

By Theorem 5.1, $D_v = r_v(R_1,\ldots,R_a) + \big[\mu_v(F_1,\ldots,F_a) - \mu_v(R_1,\ldots,R_a)\big]$, and the locally created residual satisfies

$$\|r_v(R_1,\ldots,R_a)\| \leq \rho\prod_i \|R_i\| \leq \rho M^{k_v-1}L_v$$

by Lemma A.1 and $\sum_i k_{c_i} = k_v - 1$. Adding the two contributions gives $\|D_v\| \leq k_v\rho M^{k_v-1}L_v$, which is (2).

Applying (2) at the root gives $E_T^{\mathrm{amb}} = \|D_\varrho\| \leq k\rho M^{k-1}L_T$.

For the projected error, $R = R_\varrho \in \operatorname{ran} P$ gives $PR = R$, hence $PF - R = P(F-R) = PD_\varrho$. Since $Pr_\varrho = 0$ by Definition 3.2,

$$PD_\varrho = P\big[\mu_\varrho(F_1,\ldots,F_a) - \mu_\varrho(R_1,\ldots,R_a)\big],$$

and $\|P\| = 1$ together with the bound just established for the right-hand side of (3) gives $E_T^{\mathrm{proj}} \leq (k-1)\rho M^{k-1}L_T$. Proposition 6.1 identifies E_T^{red} with E_T^{proj}.

For the orthogonal error, $(I-P)R = 0$ gives $(I-P)F = (I-P)D_\varrho$, and $\|I-P\| = 1$ gives $E_T^{\perp} \leq \|D_\varrho\| \leq k\rho M^{k-1}L_T$. □

Remark A.3. The proof uses multilinearity, the operator-norm hypotheses, contractivity of P and of $I-P$, and — for the projected estimate only — the identity $Pr_\varrho = 0$, which requires P to be the orthogonal projector appearing in Definition 3.2. Finite dimensionality is not used: the argument is valid for bounded multilinear maps between Hilbert spaces of any dimension. It is stated finite-dimensionally because everything else in this article is.

A.4. **The ordering theorem.** Let $e_i, r_i, f_i, G_i \geq 0$, put $w_i = G_i e_i$ and $d_i = f_i - r_i$, and for a permutation π of $[a]$ set

$$C(\pi) = \sum_{t=1}^{a} w_{\pi_t}\prod_{s<t} r_{\pi_s}\prod_{s>t} f_{\pi_s}.$$

Lemma A.4 (Adjacent exchange)**.** *Let π and π' agree except that π places i immediately before j and π' places j immediately before i. Then*

$$C(\pi) \leq C(\pi') \iff w_i d_j \leq w_j d_i.$$

Proof. Write A for the product of r over the slots preceding the pair and B for the product of f over the slots following it. Both are common to π and π', as is every summand indexed by a slot outside the pair. The two summands indexed by the pair contribute $AB(w_i f_j + r_i w_j)$ to $C(\pi)$ and $AB(w_j f_i + r_j w_i)$ to $C(\pi')$. Since $A, B \geq 0$, the comparison reduces to

$$w_i f_j + r_i w_j \leq w_j f_i + r_j w_i \iff w_i(f_j - r_j) \leq w_j(f_i - r_i) \iff w_i d_j \leq w_j d_i. \qquad \square$$

Lemma A.5 (The exchange relation is a total preorder)**.** *Assign to each slot the key*

$$\kappa_i = (\mathrm{cls}(i),\, q_i), \qquad \mathrm{cls}(i) = \begin{cases} 0, & d_i > 0,\\ 1, & d_i = 0,\\ 2, & d_i < 0,\end{cases} \qquad q_i = \begin{cases} w_i/d_i, & d_i \neq 0,\\ 0, & d_i = 0,\end{cases}$$

ordered lexicographically. Then $\kappa_i \leq \kappa_j$ implies $w_i d_j \leq w_j d_i$, and the lexicographic order is a total preorder on the slots.

Proof. Four cases, using $w_i, w_j \geq 0$ throughout.

Same nonzero class. If d_i, d_j are both positive or both negative then $d_i d_j > 0$, and dividing $w_i d_j \leq w_j d_i$ by $d_i d_j$ shows it is equivalent to $w_i/d_i \leq w_j/d_j$, that is, to $q_i \leq q_j$.

Positive before negative. If $d_i > 0$ and $d_j < 0$ then $w_i d_j \leq 0 \leq w_j d_i$, so the inequality holds; here $\mathrm{cls}(i) = 0 < 2 = \mathrm{cls}(j)$.

Positive before zero. If $d_i > 0$ and $d_j = 0$ then $w_i d_j = 0 \leq w_j d_i$, and $\mathrm{cls}(i) = 0 < 1 = \mathrm{cls}(j)$.

Zero before negative. If $d_i = 0$ and $d_j < 0$ then $w_i d_j \leq 0 = w_j d_i$, and $\mathrm{cls}(i) = 1 < 2 = \mathrm{cls}(j)$.

Lexicographic order on pairs drawn from totally ordered sets is a total preorder. □

Proof of Theorem 8.1. Let $\pi^\star$ sort the slots by nondecreasing κ; this is the order described in the statement. Let π be arbitrary. Because κ induces a total preorder (Lemma A.5), π can be transformed into $\pi^\star$ by a finite sequence of adjacent transpositions, each swapping a pair that occurs in the order j, i with $\kappa_i \leq \kappa_j$; this is the standard exchange-sorting argument, and it terminates because each such transposition strictly decreases the number of pairs inverted with respect to κ. By Lemmas A.5 and A.4, no transposition increases C. Hence $C(\pi^\star) \leq C(\pi)$ for every π, so $\pi^\star$ is a global minimiser. If $w_i = d_i = 0$ then the inequality of Lemma A.4 holds in both directions against every other slot, so such slots may be placed anywhere. □

Remark A.6. Theorem 8.1 minimises C, which is one particular scalar bound. It does not assert that the bound obtained with the optimal order is the smallest valid bound obtainable by any means.

A.5. **The state-resolved recursion and the pathwise expansion.**

Proof of Theorem 7.1. At a leaf all five stored quantities are exact: $\|F_v\| = \|R_v\| = \|z_v\|$ and the three error quantities vanish.

At an internal vertex, expand by Theorem 5.1 and split each child discrepancy orthogonally as $D_i = D_i^{\|} + D_i^{\perp}$. Every resulting term is $\mu_v(Z_1^\sigma, \ldots, Z_a^\sigma)$ for exactly one state vector $\sigma \in \{R, \|, \perp\}^a$. Projecting the output onto the component χ and applying the operator-norm inequality bounds the norm of that term by $\beta_v^\chi(\sigma) \prod_i \|Z_i^{\sigma_i}\|$, hence by $\beta_v^\chi(\sigma) \prod_i b_{c_i}^{\sigma_i}$ once the inductive bounds are substituted. The state vector $(R, \ldots, R)$ contributes the reduced value together with the locally created residual: its projection onto $\operatorname{ran} P_v$ is R_v itself, which is subtracted, and its orthogonal component is $r_v(R_1, \ldots, R_a)$, of norm at most $\rho_v \prod_i b_{c_i}^R$. This accounts for $\lambda_v^{\|} = 0$ and $\lambda_v^{\perp} \leq \rho_v \prod_i b_{c_i}^R$. Summing over the remaining state vectors and applying the triangle inequality gives the stated recursion. Taking, at any vertex, the minimum of several independently valid upper bounds preserves validity.

For the count: there are $3^{a_v} - 1$ state vectors other than $(R, \ldots, R)$ at v, and each contributes one product of a_v factors, so the number of combination steps at v is $O(3^{a_v})$; summing over the vertices gives $O(|T|\, 3^{a_{\max}})$. □

Proof of Corollary 10.2. Two separate assertions are proved.

Comparison. We show $B_v \leq \widehat{B}_v$ by induction on the subtree at v. At a leaf both are 0. At an internal vertex, the inductive hypothesis gives $B_{c_j} \leq \widehat{B}_{c_j}$ for every child, and the gains satisfy $h_{v,j} \geq 0$, so

$$B_v \;\leq\; \lambda_v + \sum_j h_{v,j} B_{c_j} \;\leq\; \lambda_v + \sum_j h_{v,j} \widehat{B}_{c_j} \;=\; \widehat{B}_v,$$

the first inequality being (1) and the last Definition 10.1. Nonnegativity of the gains is what makes the middle step valid, and it holds because each $h_{v,j}$ is an operator norm.

Closed form. We show $\widehat{B}_v = \sum_{u \in \mathrm{Int}\, T_v} \lambda_u \prod_{e \in \mathrm{path}(u,v)} h_e$, where T_v is the subtree rooted at v, again by induction. At a leaf both sides are 0, the sum being empty. At an internal vertex v, Definition 10.1 and the inductive hypothesis give

$$\widehat{B}_v = \lambda_v + \sum_j h_{v,j} \sum_{u \in \operatorname{Int} T_{c_j}} \lambda_u \prod_{e \in \operatorname{path}(u,c_j)} h_e = \lambda_v + \sum_j \sum_{u \in \operatorname{Int} T_{c_j}} \lambda_u \prod_{e \in \operatorname{path}(u,v)} h_e,$$

because prefixing the edge from c_j to v, whose gain is $h_{v,j}$, to the path from u to c_j gives exactly the path from u to v. The internal vertices of T_v are v together with the disjoint union of the internal vertices of the subtrees T_{c_j}, and the term for $u = v$ is λ_v with an empty product. This is the asserted sum. Taking $v = \varrho$ gives the statement, and combining the two assertions gives $B_\varrho \leq \widehat{B}_\varrho$.

Projected variant. Setting $\lambda_\varrho = 0$, which is $\lambda_\varrho^{\|} = 0$ from Theorem 7.1, removes the term $u = \varrho$ from the sum; the remaining derivation is unchanged, with the final edge of each path carrying the projected-output gain. ☐

Remark A.7. The closed form is an identity for $\widehat{B}$, and $\widehat{B}$ majorises B. It is not an identity for B, and it is not an expansion of the multilinear cross terms of Theorem 5.1: the interaction terms indexed by subsets of size at least two were absorbed into the gains $h_{v,j}$ when the scalar recurrence was formed. The result attributes a share of the bound to individual sources, and not a share of the error.

A.6. **Signed combinations and approximation of the maps.**

Proof of Corollary 11.1. Linearity of P and of the evaluation in each tree gives $PF_{\mathcal{F}} - R_{\mathcal{F}} = \sum_\alpha c_\alpha (PF_{T_\alpha} - R_{T_\alpha})$, and the triangle inequality bounds the norm by $\sum_\alpha |c_\alpha| B^{\mathrm{proj}}_{T_\alpha}$. For the five-input ternary associator, each of the two trees has two internal vertices, so Theorem 9.1 gives $B^{\mathrm{proj}}_{T_\alpha} \leq (2-1)\rho M L = \rho M L$ and the two terms give $2\rho M L$; the ambient bound is $2 \cdot 2\rho M L = 4\rho M L$. Syntactically identical trees may be combined exactly before this estimate is applied. ☐

Lemma A.8 (Sensitivity of the projected evaluation to the maps)**.** *Let $\{\mu_v\}$ and $\{\widehat{\mu}_v\}$ satisfy $\|\mu_v\|_{\mathrm{op}} \leq M$ and $\|\mu_v - \widehat{\mu}_v\|_{\mathrm{op}} \leq \delta$ at every internal vertex, and let R_μ and $R_{\widehat{\mu}}$ be the recursively projected evaluations of the two trees on the same leaf data. Then for every subtree rooted at v,*

$$\|R^\mu_v - R^{\widehat{\mu}}_v\| \leq k_v\, \delta\, (M+\delta)^{k_v - 1} L_v \quad (k_v \geq 1), \qquad \|R^\mu_v - R^{\widehat{\mu}}_v\| = 0 \quad (k_v = 0).$$

Proof. At a leaf both evaluations equal $Q_{\tau(v)} z_v$, so the difference vanishes. At an internal vertex v of arity a, using $\|P_v\| = 1$ and adding and subtracting $\widehat{\mu}_v(R^\mu_{c_1}, \dots, R^\mu_{c_a})$,

$$\begin{aligned}\|R^\mu_v - R^{\widehat{\mu}}_v\| &= \big\| P_v \big[\mu_v(R^\mu_{c_\bullet}) - \widehat{\mu}_v(R^{\widehat{\mu}}_{c_\bullet})\big] \big\| \\ &\leq \underbrace{\big\|(\mu_v - \widehat{\mu}_v)(R^\mu_{c_1}, \dots, R^\mu_{c_a})\big\|}_{\text{(I)}} + \underbrace{\big\|\widehat{\mu}_v(R^\mu_{c_\bullet}) - \widehat{\mu}_v(R^{\widehat{\mu}}_{c_\bullet})\big\|}_{\text{(II)}}.\end{aligned}$$

For (I), Lemma A.1 gives $\|R^\mu_{c_i}\| \leq M^{k_{c_i}} L_{c_i}$, so (I) $\leq \delta M^{k_v - 1} L_v \leq \delta (M+\delta)^{k_v - 1} L_v$. For (II), telescoping the arguments of $\widehat{\mu}_v$ one at a time and using $\|\widehat{\mu}_v\|_{\mathrm{op}} \leq M + \delta$ together with the magnitude bounds $\|R^\mu_{c_i}\|, \|R^{\widehat{\mu}}_{c_i}\| \leq (M+\delta)^{k_{c_i}} L_{c_i}$ from Lemma A.1 applied with $M + \delta$ in place of M,

$$\text{(II)} \leq (M+\delta) \sum_j \|R^\mu_{c_j} - R^{\widehat{\mu}}_{c_j}\| \prod_{i \neq j} (M+\delta)^{k_{c_i}} L_{c_i} \leq \Big(\sum_j k_{c_j}\Big) \delta (M+\delta)^{k_v - 1} L_v,$$

by the inductive hypothesis, the summand with $k_{c_j} = 0$ vanishing. Since $\sum_j k_{c_j} = k_v - 1$, the two contributions add to $k_v \delta (M+\delta)^{k_v - 1} L_v$. ☐

Proof of Proposition 14.1. Insert the recursively projected *exact* tree:

$$\|F_\mu - R_{\widehat{\mu}}\| \leq \|F_\mu - R_\mu\| + \|R_\mu - R_{\widehat{\mu}}\|.$$

The first term is E_T^{amb} for the exact tree, bounded by $k\rho M^{k-1}L_T$ by Theorem 9.1; the second is bounded by $k\delta(M+\delta)^{k-1}L_T$ by Lemma A.8. This gives the first estimate.

For the second, $R_\mu \in \operatorname{ran} P$ gives $PR_\mu = R_\mu$, so

$$\|PF_\mu - R_{\widehat{\mu}}\| \leq \|PF_\mu - R_\mu\| + \|R_\mu - R_{\widehat{\mu}}\|,$$

and the first term is E_T^{proj} for the exact tree, bounded by $(k-1)\rho M^{k-1}L_T$ by Theorem 9.1.

Both estimates use only the declared hypotheses: $\|\mu_v\|_{\mathrm{op}} \leq M$ and $\|r_v\|_{\mathrm{op}} \leq \rho$ for the exact maps, and $\|\mu_v - \widehat{\mu}_v\|_{\mathrm{op}} \leq \delta$. In particular no bound is assumed on $(I-P_v)\widehat{\mu}_v(P\,\cdot,\ldots,P\,\cdot)$; compare Remark 14.2. No equality with the observed total error is asserted. $\square$

Appendix B. Reproducibility

All computations were carried out inside a single repository. Each run directory records the resolved configuration, the command, the environment, the hardware, the source commit, hashes of the mathematical objects and of the inputs, the tensors, the metrics, the enclosures, the logs and hashes of every output. The companion software article describes the interfaces and commands.

Remark B.1 (Provenance of the numerical evidence). The tables and figures of this article were produced at source commit `b718f4e51785`, which is earlier than the commit carrying the text. The classification of enclosure outcomes in Table 10 was re-derived from the certified lower and upper bounds recorded at that commit, without re-running any experiment; the re-derivation script and the reclassified registry accompany this article. Full details are in the provenance file of the delivery package.

TABLE 1. Overlap with the cited literature, by area. A bounded search cannot establish originality; see Remark 2.1.

area	overlap	difference	comparison with the cited result
Yau, Colored Operads	The type grammar and ordered tree model.	Orthogonal projection defects and quantitative vertexwise bounds are additional analytic data.	standard restriction
Loday and Vallette, Algebraic Operads	Signed forests and insertion-tree syntax.	Quantitative orthogonal leakage is not the monograph's central problem.	specialisation of a known bound
Johnson, Approximately multiplicative maps between Banach algebras	Quantitative stability language for algebraic structure.	The present work propagates vertexwise orthogonal projection defects on a fixed finite tree, rather than correcting a map globally.	originality not assessed
Higham, Accuracy and Stability of Numerical Algorithms	Local error propagation and evaluation-order comparisons.	Exact multilinear projector residuals and typed tree topology.	specialisation of a known bound
Combettes and Pesquet, Lipschitz bounds for layered network structures	Recursive propagation of local operator summaries.	The present work uses multilinear branching, orthogonal and projected blocks, and locally created residuals.	originality not assessed
Gehr et al., AI2	Vertexwise abstract summaries and recursively propagated enclosures.	No neural activation or robustness property; exact multilinear projector decomposition.	originality not assessed
Hackbusch and Kuehn, A New Scheme for the Tensor Representation	Topology-dependent finite tree calculations.	The present work studies recursive projection of the outputs of multilinear maps, rather than rank truncation of a single tensor.	originality not assessed
Ballani and Grasedyck, Tree Adaptive Approximation in the Hierarchical Tensor Format	Tree topology as an experimental factor.	Constants concern closure and recursive projection, not hierarchical ranks.	originality not assessed
Kolda and Bader, Tensor Decompositions and Applications	Secondary CP representation budget.	CP error is separated from closure error.	specialisation of a known bound
Moore, Kearfott, and Cloud, Introduction to Interval Analysis	Small explicit lower-construction enclosures.	Application to tree constant ratios only.	standard multilinear bound
Lasserre, Global Optimization with Polynomials and the Problem of Moments	Global bounds for small cases.	No semidefinite relaxation is reported as a bound here, since solver and relaxation status were not independently validated.	standard multilinear bound
Egger et al., Structure-Preserving Model Reduction	Projection hypotheses and quantified loss of structure.	Static finite multilinear composition trees and closure leakage.	originality not assessed

Table 2. Notation.

symbol	generated meaning
V_τ, W_τ	ambient and reduced Hilbert spaces of type τ
Q_τ, P_τ	isometry and orthogonal projector $P_\tau = Q_\tau Q_\tau^*$
F_v, R_v	ambient and recursively projected values at node v
$D_v^{\parallel}, D_v^{\perp}$	projected and orthogonal components of $F_v - R_v$
M_v, ρ_v	local operator-norm and closure-residual bounds
B_v^F, B_v^R	bounds for $\|F_v\|$ and $\|R_v\|$
$B_v^{\mathrm{amb}}, B_v^{\parallel}, B_v^{\perp}$	ambient, projected, and orthogonal error bounds
$C_T^{\mathrm{amb}}, C_T^{\mathrm{proj}}$	best universal constants for a typed tree T

Table 3. Vertexwise formulas. Each row is an upper-bound contract, not an equality with the observed error.

bounds	generated formula	status
homogeneous ambient	$k\rho M^{k-1}\prod_\ell \|z_\ell\|$	proved
homogeneous projected	$(k-1)\rho M^{k-1}\prod_\ell \|z_\ell\|$	proved
nodewise subset	$\rho_v \prod_i B_i^R + M_v \sum_{\varnothing\neq S}\prod_{i\in S} B_i^A \prod_{j\notin S} B_j^R$	proved
pathwise	$\sum_v \rho_v \prod_{e\in\pi(v)} L_e$	proved upper
optimized telescoping	$\min_\pi \sum_j w_{\pi_j}(f_{\pi_j} - r_{\pi_j})\prod_{i<j} f_{\pi_i}\prod_{i>j} r_{\pi_i}$	proved upper

Table 4. Comparison of the four upper bounds, as median ratios to the uniform ambient bound on the same configurations.

local pattern	nodewise	pathwise	state resolved	optimized	instances
alternating	0.213	0.213	0.355	0.213	540
deepest_path	0.667	0.667	0.668	0.667	540
leaves	0.417	0.417	0.692	0.417	540
one_dominant	0.167	0.167	0.168	0.167	540
random_log_uniform	0.016	0.016	0.022	0.016	540
root	0.167	0.167	0.168	0.167	540
two_branches	0.667	0.667	0.715	0.667	540
uniform	1.000	1.000	1.642	1.000	540

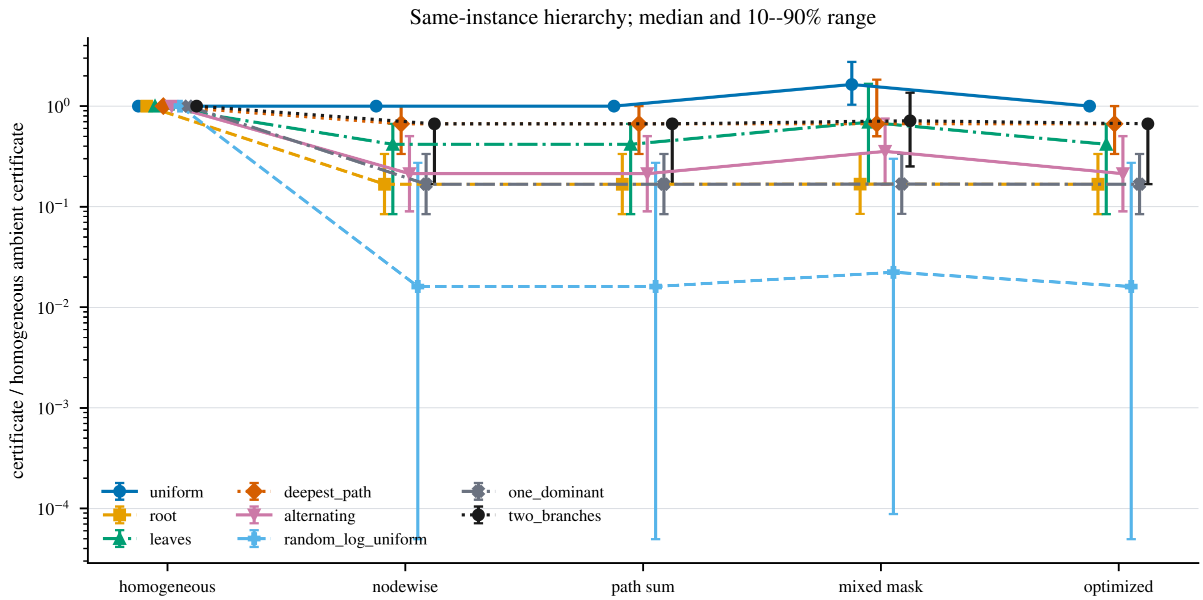

FIGURE 3. Comparison of four upper bounds on 4,320 distinct configurations with heterogeneous local parameters (10 declared parameter draws; no optimiser restarts are involved, since all four bounds are computed in closed form). Each bound is divided by the uniform ambient bound of Theorem 9.1 on the same configuration; points are medians and bars are the 10th–90th percentile range over configurations, not an uncertainty estimate. The data show that the vertexwise, pathwise and order-optimised bounds improve on the uniform bound when the local parameters are nonuniform. They do not show that any of these bounds is optimal.

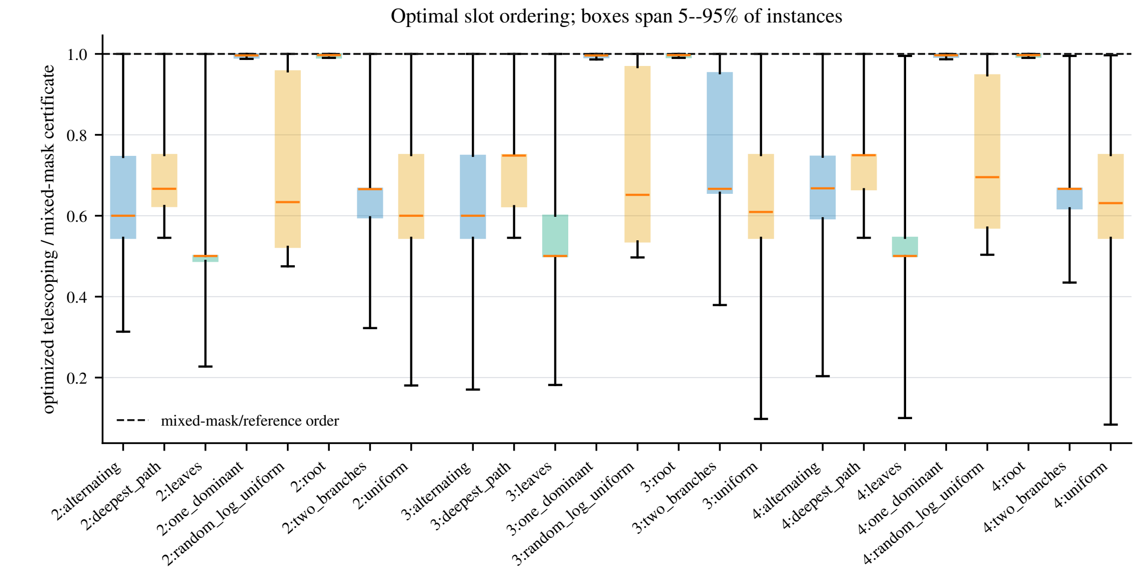

Figure 4. Effect of the ordering rule of Theorem 8.1, over all 4,320 distinct configurations. The vertical axis is the bound obtained with the optimal order divided by the bound obtained with the reference order. Boxes give the median, quartiles and 5th–95th percentile range over configurations. The rule is never worse and is sometimes strictly better. It is optimal within the scalar telescoping family of Section 8 only, not among all bounding strategies.

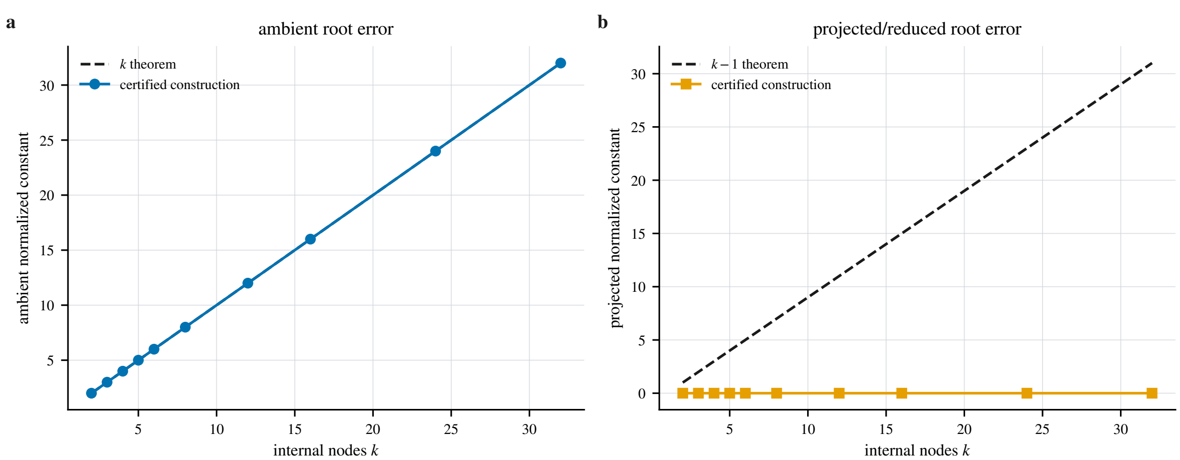

Figure 5. The coefficients k and $k-1$ on 10 binary left-comb trees in dimension two at the smallest tested η, with interval-enclosed lower constructions. Ambient and projected errors are normalised by ηM^{k-1} and compared with the proved coefficients. The ambient construction attains the linear coefficient in these configurations. The projected construction shown here, at small η, remains far below $k-1$; whether $k-1$ is attained at fixed $\eta > 0$ is open.

TABLE 5. Uniform ambient constants. Lower values are interval-enclosed values of explicit constructions.

arity	k	best ambient lower	proved upper bound	lower/upper
2	2	2.000	2.000	1.000
2	3	3.000	3.000	1.000
2	4	4.000	4.000	1.000
2	5	5.000	5.000	1.000
2	6	6.000	6.000	1.000
2	8	8.000	8.000	1.000
2	12	12.000	12.000	1.000
2	16	16.000	16.000	1.000
2	24	24.000	24.000	1.000
2	32	32.000	32.000	1.000
3	2	2.000	2.000	1.000
3	3	3.000	3.000	1.000
3	4	4.000	4.000	1.000
3	5	5.000	5.000	1.000
3	6	6.000	6.000	1.000
3	8	8.000	8.000	1.000
3	12	12.000	12.000	1.000
3	16	16.000	16.000	1.000
3	24	24.000	24.000	1.000
3	32	32.000	32.000	1.000
4	2	2.000	2.000	1.000
4	3	3.000	3.000	1.000
4	4	4.000	4.000	1.000
4	5	5.000	5.000	1.000
4	6	6.000	6.000	1.000
4	8	8.000	8.000	1.000
4	12	12.000	12.000	1.000
4	16	16.000	16.000	1.000
4	24	24.000	24.000	1.000
4	32	32.000	32.000	1.000

Table 6. Uniform projected-error constants. At small η the explicit lower constructions remain far below the proved upper bound.

arity	k	best projected-error lower bound	proved upper bound	lower/upper
2	2	1.00×10^{-8}	1.000	1.00×10^{-8}
2	3	3.00×10^{-8}	2.000	1.50×10^{-8}
2	4	6.00×10^{-8}	3.000	2.00×10^{-8}
2	5	1.00×10^{-7}	4.000	2.50×10^{-8}
2	6	1.50×10^{-7}	5.000	3.00×10^{-8}
2	8	2.80×10^{-7}	7.000	4.00×10^{-8}
2	12	6.60×10^{-7}	11.000	6.00×10^{-8}
2	16	1.20×10^{-6}	15.000	8.00×10^{-8}
2	24	2.76×10^{-6}	23.000	1.20×10^{-7}
2	32	4.96×10^{-6}	31.000	1.60×10^{-7}
3	2	1.00×10^{-8}	1.000	1.00×10^{-8}
3	3	3.00×10^{-8}	2.000	1.50×10^{-8}
3	4	6.00×10^{-8}	3.000	2.00×10^{-8}
3	5	1.00×10^{-7}	4.000	2.50×10^{-8}
3	6	1.50×10^{-7}	5.000	3.00×10^{-8}
3	8	2.80×10^{-7}	7.000	4.00×10^{-8}
3	12	6.60×10^{-7}	11.000	6.00×10^{-8}
3	16	1.20×10^{-6}	15.000	8.00×10^{-8}
3	24	2.76×10^{-6}	23.000	1.20×10^{-7}
3	32	4.96×10^{-6}	31.000	1.60×10^{-7}
4	2	1.00×10^{-8}	1.000	1.00×10^{-8}
4	3	3.00×10^{-8}	2.000	1.50×10^{-8}
4	4	6.00×10^{-8}	3.000	2.00×10^{-8}
4	5	1.00×10^{-7}	4.000	2.50×10^{-8}
4	6	1.50×10^{-7}	5.000	3.00×10^{-8}
4	8	2.80×10^{-7}	7.000	4.00×10^{-8}
4	12	6.60×10^{-7}	11.000	6.00×10^{-8}
4	16	1.20×10^{-6}	15.000	8.00×10^{-8}
4	24	2.76×10^{-6}	23.000	1.20×10^{-7}
4	32	4.96×10^{-6}	31.000	1.60×10^{-7}

TABLE 7. Largest values found by numerical search, against the triangle bound of Corollary 11.1. These are lower bounds obtained by optimisation. An optimiser can improve a lower bound; it cannot establish that no larger value exists.

combination	triangle bound	best value	ratio	status
five-input ternary associator	2.0	1.373	0.686	undetermined
anchored associator	2.0	1.658	0.829	undetermined
Jacobi combination (three terms)	3.0	2.983	0.994	undetermined
Filippov fundamental identity	4.0	1.664	0.416	widest gap
six-term generalised Jacobi expression	6.0	$\sim$0	$\sim$0	see §11.2

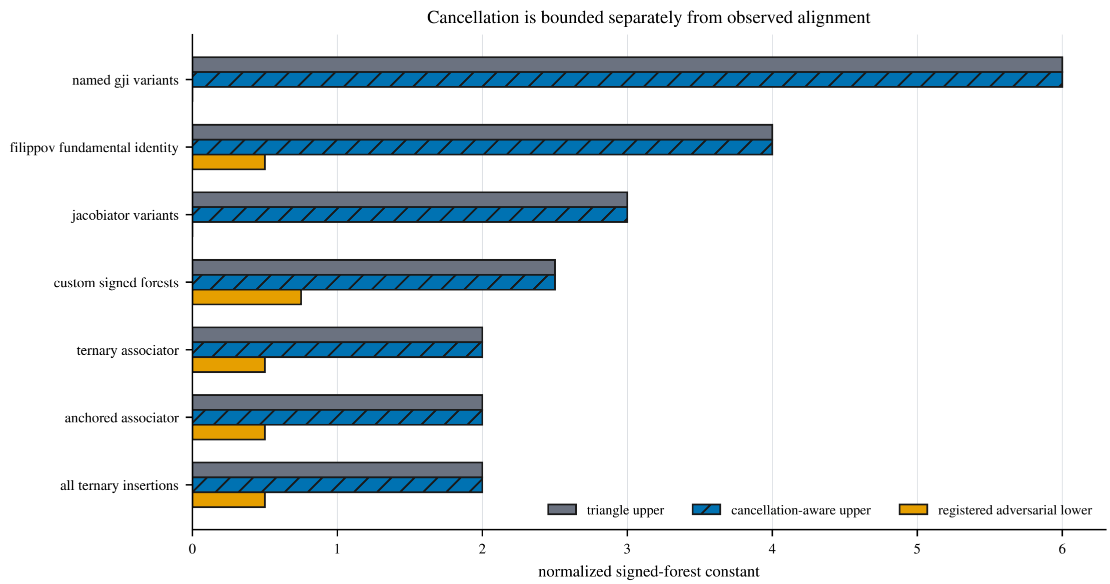

FIGURE 6. Signed linear combinations of composition trees: seven families over 28 configurations. The exact quantity plotted is the normalised residual of the signed combination. The two upper bars are proved upper bounds — the triangle bound and the bound after exact combination of syntactically identical trees — while the third bar is the largest lower value found by numerical search over η. The comparison shows that exact combination improves the bookkeeping for these declared expressions. The search values are lower bounds and do not determine the constants for the associator, the Filippov identity, or the generalised Jacobi expressions.

TABLE 8. Constants for associator and signed-combination families.

expression	triangle	cancellation upper	observed	observed/triangle	search status
all_ternary_insertions	2.000	2.000	0.500	0.250	not executed, under a stated computational-cost constraint
anchored_associator	2.000	2.000	0.500	0.250	not executed, under a stated computational-cost constraint
custom_signed_forests	2.500	2.500	0.750	0.300	not executed, under a stated computational-cost constraint
filippov_fundamental_identity	4.000	4.000	0.500	0.125	not executed, under a stated computational-cost constraint
five_input_ternary_associator	2.000	2.000	0.500	0.250	not executed, under a stated computational-cost constraint
jacobiator_variants	3.000	3.000	0	0	not executed, under a stated computational-cost constraint
named_gji_variants	6.000	6.000	2.22×10^{-12}	3.70×10^{-13}	not executed, under a stated computational-cost constraint

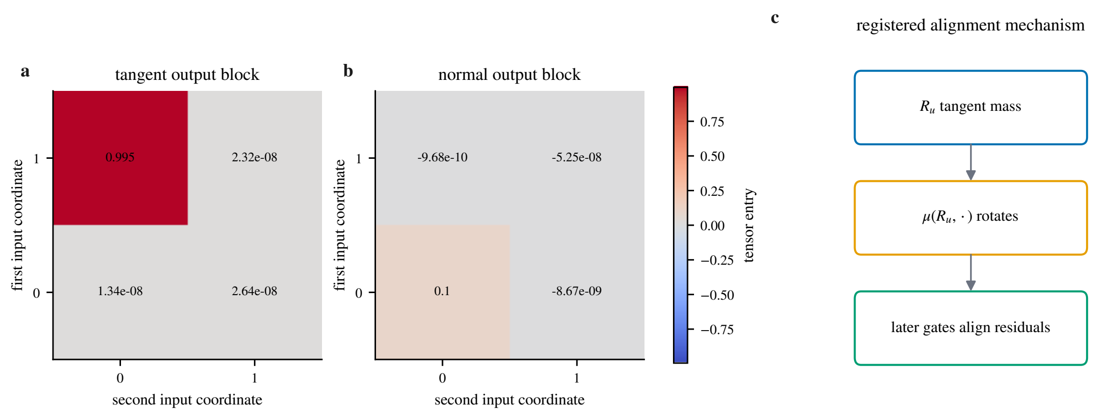

FIGURE 7. Geometry of the extremal construction. Panels (a) and (b) display the projected and orthogonal output blocks of one explicit rank-one gated rotation tensor; panel (c) summarises the mechanism by which an orthogonal error created at a lower vertex acquires a projected component at its parent. Entries are the exact stored inputs up to floating-point representation. The figure establishes admissibility and alignment of this construction, not its uniqueness or global optimality.

TABLE 9. Best explicit constructions by error type.

error	construction	arity	k	η	lower/upper	status
ambient	gated planar rotation	2	1	0.500	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP
normal	gated planar rotation	2	1	0.500	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP
projected	gated planar rotation	2	2	1.000	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP

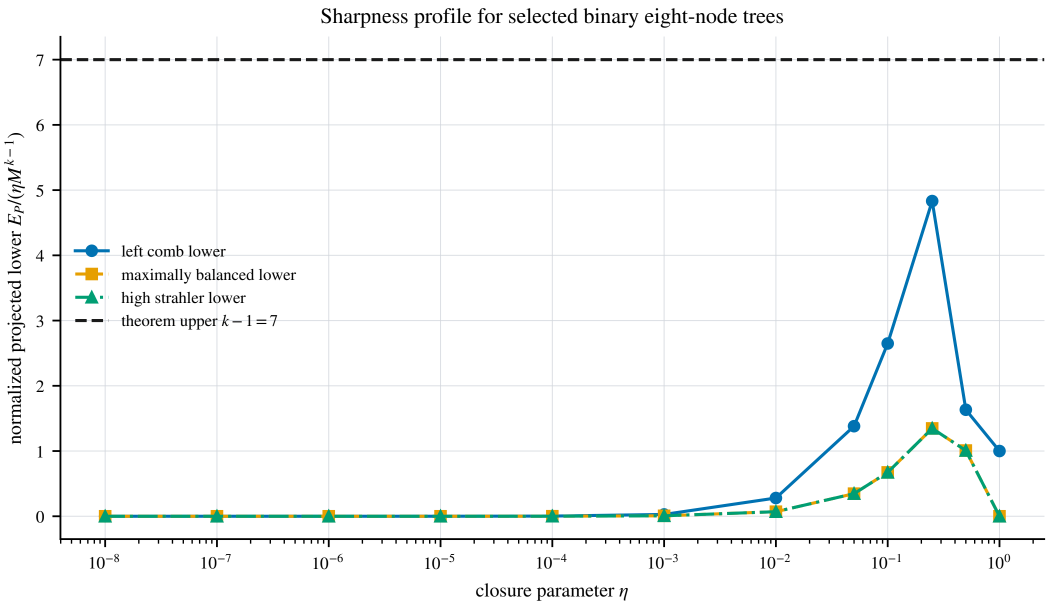

FIGURE 8. Explicit lower constructions for the projected-error constant on 96 distinct binary trees with eight internal vertices. The vertical axis is the interval-enclosed projected error divided by ηM^{k-1}; the dashed line is the proved upper coefficient $k-1=7$. The construction is explicit, so there is no statistical uncertainty. The figure exhibits an admissible lower profile; it does not determine $C_T^{\mathrm{proj}}(\eta)$.

TABLE 10. Outcome of comparing the interval-enclosed lower construction with the proved upper bound, by error type. *Determined*: the two bounds coincide at a positive value. *Zero*: the proved upper bound is itself zero, so the constant vanishes as an immediate consequence of Theorem 9.1. *Partial*: a positive lower bound was obtained that does not meet the upper bound. *None*: no positive lower bound was obtained, so the only lower bound available is the trivial one implied by nonnegativity. The four columns are mutually exclusive and sum to the configuration count.

		outcome				
error type	#	determined	zero	partial	none	median gap
ambient	4275	258	0	3622	395	3
normal	1395	30	0	1098	267	2.701
projected	4275	21	30	3092	1132	3
all	9945	309	30	7812	1794	3

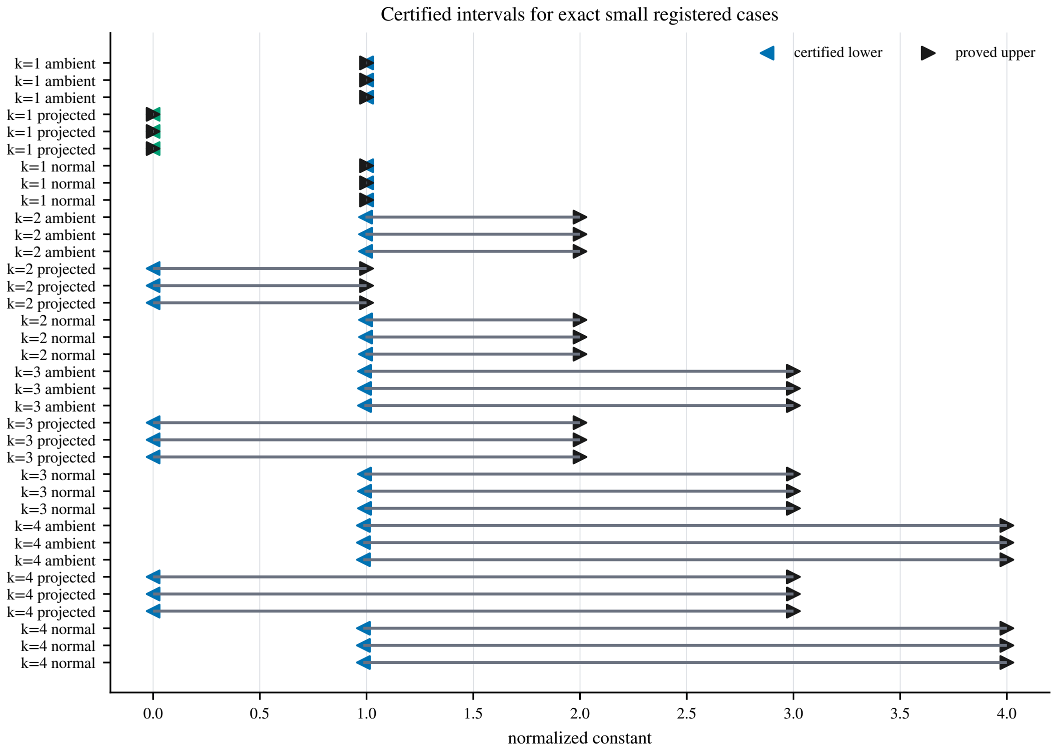

FIGURE 9. Enclosures of the constant for 36 small binary configurations at $\eta = 0.1$. Each horizontal interval joins the interval-enclosed lower construction to the proved upper bound; a filled marker identifies a configuration in which the two coincide. The references are the coefficients k and $k-1$. Only the marked configurations have a determined constant; the remaining intervals are open, and their width is not an error estimate.

TABLE 11. Small configurations in which the constant is determined.

arity	k	error	lower	upper	global bounds
2	1	ambient	1.000	1.000	yes
2	1	orthogonal	1.000	1.000	yes
2	1	projected	0	0	yes
2	2	ambient	1.000	2.000	no
2	2	orthogonal	1.000	2.000	no
2	2	projected	0	1.000	no
2	3	ambient	1.000	3.000	no
2	3	orthogonal	1.000	3.000	no
2	3	projected	0	2.000	no
2	4	ambient	1.000	4.000	no
2	4	orthogonal	1.000	4.000	no
2	4	projected	0	3.000	no
3	1	ambient	1.000	1.000	yes
3	1	orthogonal	1.000	1.000	yes
3	1	projected	0	0	yes
3	2	ambient	1.000	2.000	no
3	2	orthogonal	1.000	2.000	no
3	2	projected	0	1.000	no
3	3	ambient	1.000	3.000	no
3	3	orthogonal	1.000	3.000	no
3	3	projected	0	2.000	no
3	4	ambient	1.000	4.000	no
3	4	orthogonal	1.000	4.000	no
3	4	projected	0	3.000	no

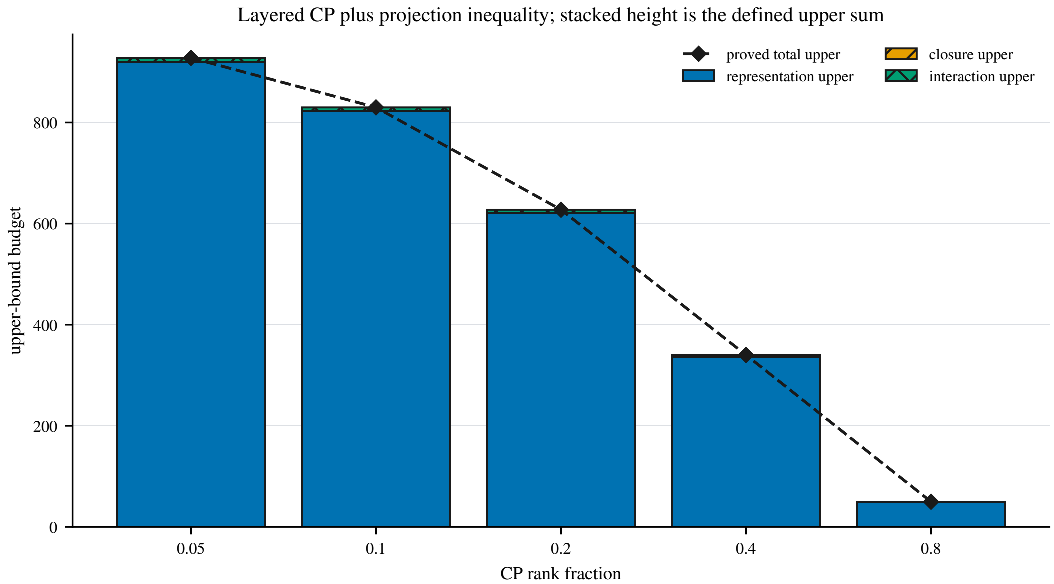

FIGURE 10. Separation of representation error from recursive-projection error, for five configurations in dimension 64 on a balanced tree with eight internal vertices at $\eta = 0.01$. The three bars are the separately proved representation, closure and interaction terms of Proposition 14.1. Their sum is by definition the proved upper bound and is not asserted to equal the observed error. The figure displays a decomposition of a bound; it says nothing about tightness or about identifiability of the low-rank representation.

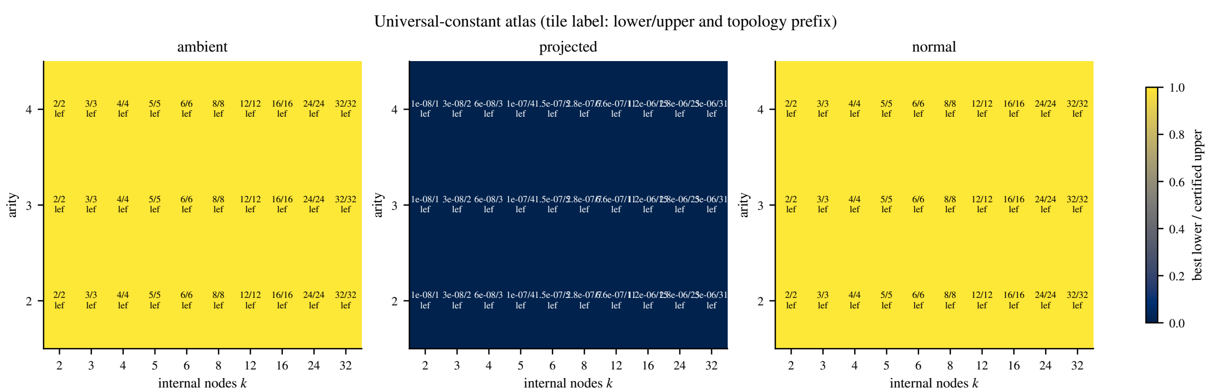

FIGURE 11. Ratios of the best explicit lower construction to the proved upper bound, over 240 two-dimensional configurations selected from 2,880. Tile colour is the ratio; the text gives the enclosed lower value, the upper bound, and the topology attaining the maximum. Lower values are rigorous interval enclosures of an explicit construction, so they carry no sampling uncertainty. A ratio of one identifies a configuration in which the constant is determined; ratios below one leave the exact constant undetermined.

TABLE 12. Representation and projection budget. The components sum to a proved upper bound, not to the observed error.

dim.	rank frac.	rank	tensor err.	representation	closure	interaction	total upper
16	0.050	1	0.968	457.967	0.050	4.070	462.087
16	0.100	2	0.935	394.174	0.050	3.619	397.843
16	0.200	3	0.901	336.334	0.050	3.198	339.582
16	0.400	6	0.791	195.688	0.050	2.102	197.840
16	0.800	13	0.433	24.042	0.050	0.428	24.521
32	0.050	2	0.968	457.967	0.050	4.070	462.087
32	0.100	3	0.952	425.309	0.050	3.841	429.199
32	0.200	6	0.901	336.334	0.050	3.198	339.582
32	0.400	13	0.771	176.675	0.050	1.942	178.667
32	0.800	26	0.433	24.042	0.050	0.428	24.521
64	0.050	3	0.976	474.879	0.050	4.187	479.116
64	0.100	6	0.952	425.309	0.050	3.841	429.199
64	0.200	13	0.893	322.770	0.050	3.097	325.917
64	0.400	26	0.771	176.675	0.050	1.942	178.667
64	0.800	51	0.451	27.128	0.050	0.469	27.648
128	0.050	6	0.976	474.879	0.050	4.187	479.116
128	0.100	13	0.948	417.384	0.050	3.785	421.218
128	0.200	26	0.893	322.770	0.050	3.097	325.917
128	0.400	51	0.776	181.317	0.050	1.982	183.348
128	0.800	102	0.451	27.128	0.050	0.469	27.648

TABLE 13. Exhaustive ordered-tree enumeration statistics.

grammar	occurrences	unique hashes	internal nodes	max depth
binary	2055	2055	1–8	8
mixed_2_3_4	78879	78879	1–5	5
quaternary	167	167	1–4	4
ternary	344	344	1–5	5

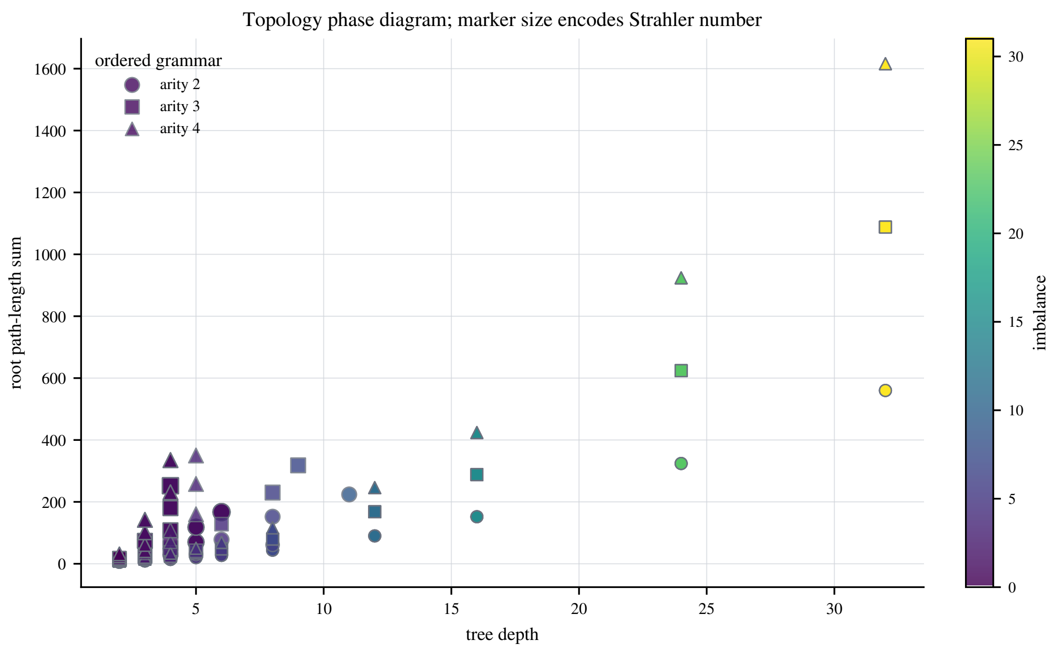

FIGURE 12. Dependence of the bound on tree topology, over 240 distinct configurations at $\eta = 0.1$. The axes are tree depth and the sum of root-to-vertex path lengths; colour encodes imbalance, marker size the Strahler number, and a highlighted edge marks a projected lower-to-upper ratio above 0.9. Topology statistics are exact and lower values are interval-enclosed. The figure shows that the bound depends on topology; it does not identify a transition or an extremal topology.

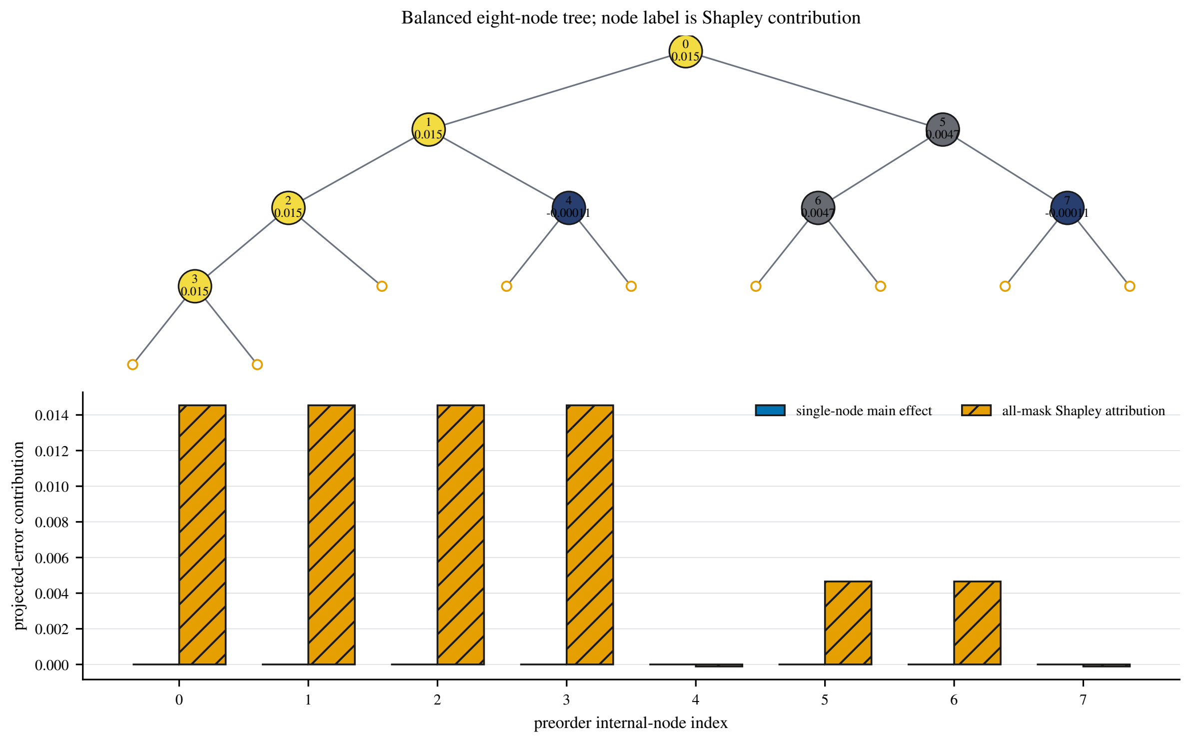

FIGURE 13. Attribution of the projected root error to individual vertices, for one balanced tree with eight internal vertices. All $2^8 = 256$ subsets of vertices at which the closure residual is switched on were evaluated exhaustively; no sampling is involved. Vertex labels give the Shapley value of the resulting set function, and the lower panel compares single-vertex main effects with the full Shapley values. The efficiency residual is below 10^{-12}. The attribution is exact for this construction and is not a claim about a unique causal mechanism.

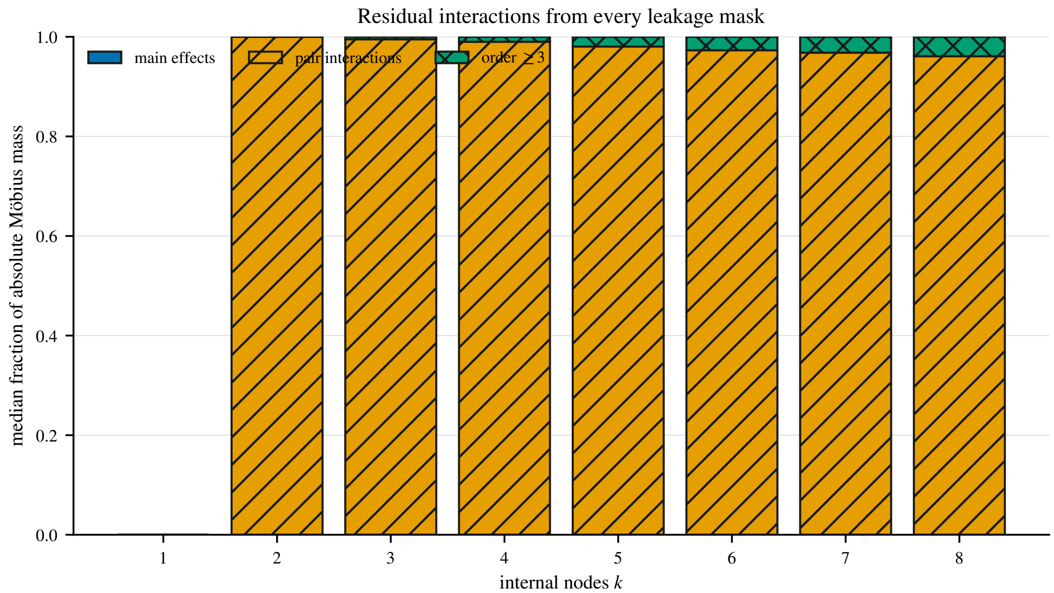

Figure 14. Interaction structure of the local error sources, from all 1,530 exhaustive subset evaluations over 24 trees. For each tree, the exact set function of the projected error is Möbius-inverted; bars give the topology-median fraction of absolute coefficient mass carried by first-, second- and higher-order subsets. The reference is the exact subset expansion of Theorem 5.1. The data show that the local error sources do not combine additively in these constructions.

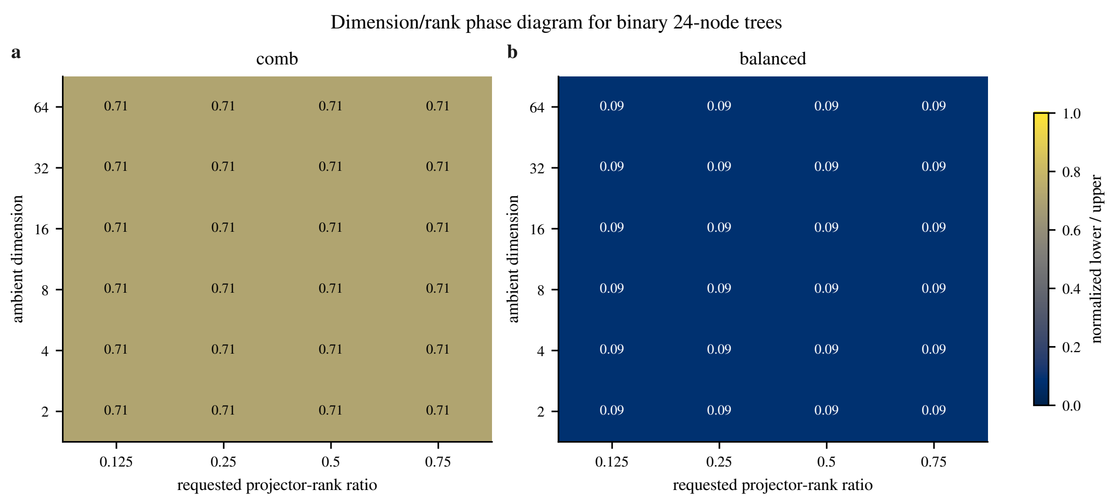

Figure 15. Dependence on ambient dimension and projector rank, over 72 configurations on binary trees with 24 internal vertices at $\eta = 0.1$. Each tile is the value of the two-dimensional construction, embedded isometrically into the stated dimension, divided by the proved upper coefficient $k-1$. Values are interval-enclosed. The embedding preserves the value, so the construction is admissible in every dimension shown. No configuration exhibits an improvement obtained from the additional coordinates.

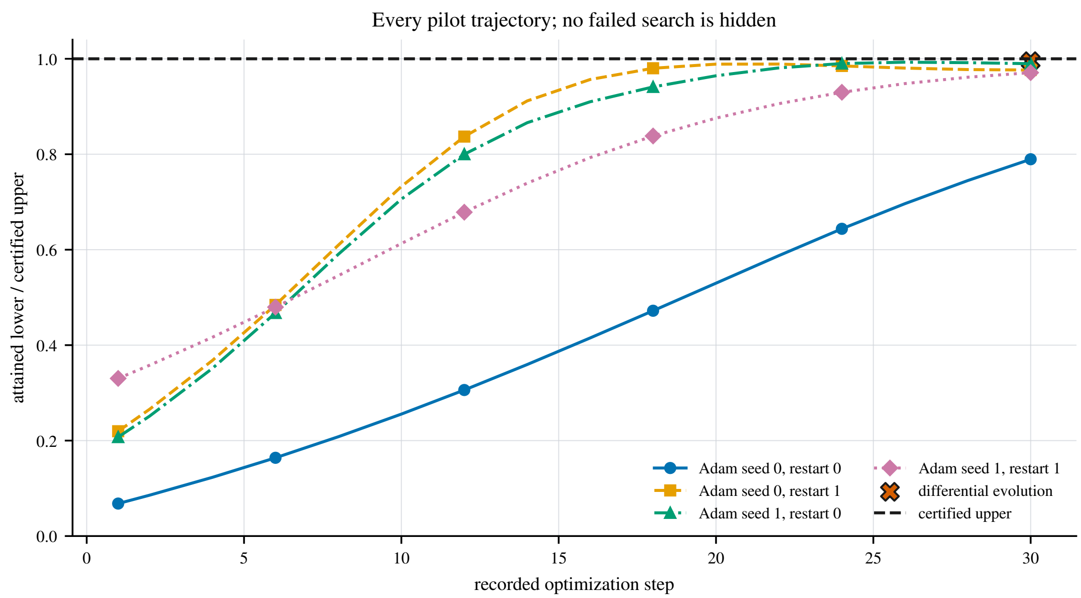

FIGURE 16. Convergence of the numerical search on a single configuration: four gradient-based trajectories and one independent derivative-free run. The vertical axis is the attained normalised lower value divided by the proved upper bound. Every recorded step of every run is shown. Optimiser restarts are not statistical replicates and no interval is inferred from them. Agreement close to one indicates a strong lower construction for this configuration; it is not a certificate of global optimality.

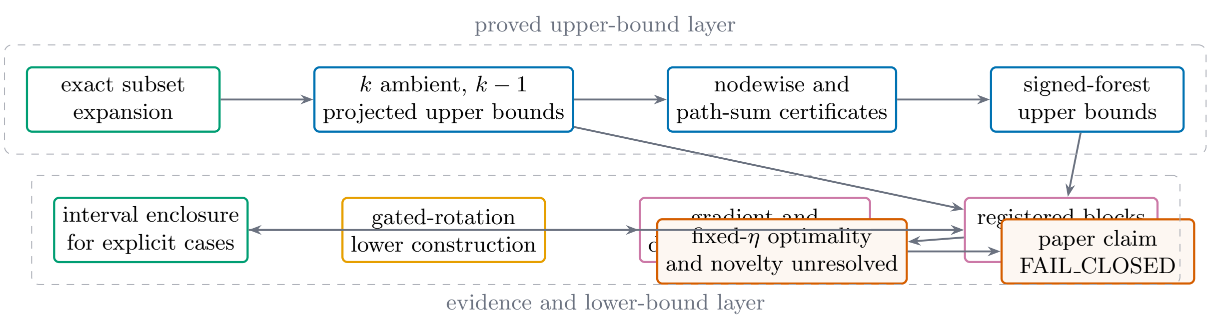

FIGURE 17. Dependency graph relating the statements of this article to their supporting material. Proved upper bounds, rigorous lower enclosures, exploratory numerical evidence, and unresolved questions are kept in separate layers. The graph is a navigation aid for the reproducibility appendix; it does not convert a numerical search result into a theorem.

Table 14. Dependence on ambient dimension and projector rank. The lower value is the two-dimensional construction embedded isometrically.

dimension	rank ratio	rank	best normalized lower	upper	improvement established
2	0.125	1	16.265	2.000	no
2	0.250	1	16.265	2.000	no
2	0.500	1	16.265	2.000	no
2	0.750	1	16.265	2.000	no
4	0.125	1	16.265	2.000	no
4	0.250	1	16.265	2.000	no
4	0.500	2	16.265	2.000	no
4	0.750	3	16.265	2.000	no
8	0.125	1	16.265	2.000	no
8	0.250	2	16.265	2.000	no
8	0.500	4	16.265	2.000	no
8	0.750	6	16.265	2.000	no
16	0.125	2	16.265	2.000	no
16	0.250	4	16.265	2.000	no
16	0.500	8	16.265	2.000	no
16	0.750	12	16.265	2.000	no
32	0.125	4	16.265	2.000	no
32	0.250	8	16.265	2.000	no
32	0.500	16	16.265	2.000	no
32	0.750	24	16.265	2.000	no
64	0.125	8	16.265	2.000	no
64	0.250	16	16.265	2.000	no
64	0.500	32	16.265	2.000	no
64	0.750	48	16.265	2.000	no

Table 15. Precision and conditioning controls.

precision	case	max residual	backward error	condition est.	min bound margin
arbitrary_precision	ill_conditioned	0	0	1.00×10^{8}	3.00×10^{-8}
arbitrary_precision	near_bound	0	0	1,000.000	0.003
arbitrary_precision	optimizer_stability	2.22×10^{-16}	2.09×10^{-16}	2.000	0.438
arbitrary_precision	small_exact	0	0	10.000	0.241
complex128	ill_conditioned	0	0	1.00×10^{8}	3.00×10^{-8}
complex128	near_bound	0	0	1,000.000	0.003
complex128	optimizer_stability	0	0	2.000	0.438
complex128	small_exact	0	0	10.000	0.241
complex64	ill_conditioned	0	0	1.00×10^{8}	3.00×10^{-8}
complex64	near_bound	8.48×10^{-13}	8.48×10^{-13}	1,000.000	0.003
complex64	optimizer_stability	4.04×10^{-8}	3.80×10^{-8}	2.000	0.438
complex64	small_exact	6.99×10^{-10}	6.99×10^{-10}	10.000	0.241
exact_rational	ill_conditioned	0	0	1.667	0.547
exact_rational	near_bound	0	0	1.667	0.547
exact_rational	optimizer_stability	0	0	1.667	0.547
exact_rational	small_exact	0	0	1.667	0.547
float32	ill_conditioned	0	0	1.00×10^{8}	3.00×10^{-8}
float32	near_bound	8.48×10^{-13}	8.48×10^{-13}	1,000.000	0.003
float32	optimizer_stability	4.04×10^{-8}	3.80×10^{-8}	2.000	0.438
float32	small_exact	6.99×10^{-10}	6.99×10^{-10}	10.000	0.241
float64	ill_conditioned	0	0	1.00×10^{8}	3.00×10^{-8}
float64	near_bound	0	0	1,000.000	0.003
float64	optimizer_stability	0	0	2.000	0.438
float64	small_exact	0	0	10.000	0.241

Table 16. Agreement between the CPU and GPU evaluators. The tolerance is a validation threshold, not a bound in any theorem.

precision	cases	max abs. difference	median difference	result
complex128	4	0	0	pass
complex64	4	1.92×10^{-8}	5.60×10^{-9}	pass
float32	4	1.92×10^{-8}	5.60×10^{-9}	pass
float64	4	0	0	pass

Table 17. Controls and unresolved items.

item	count	disposition
run failures	0	none
invalid typed edges	48	rejected before evaluation
zero projected one-node constant	30	exact control
fixed-η universal sharpness	1	open
extended optimizer grid	1	not executed, under a stated cost constraint

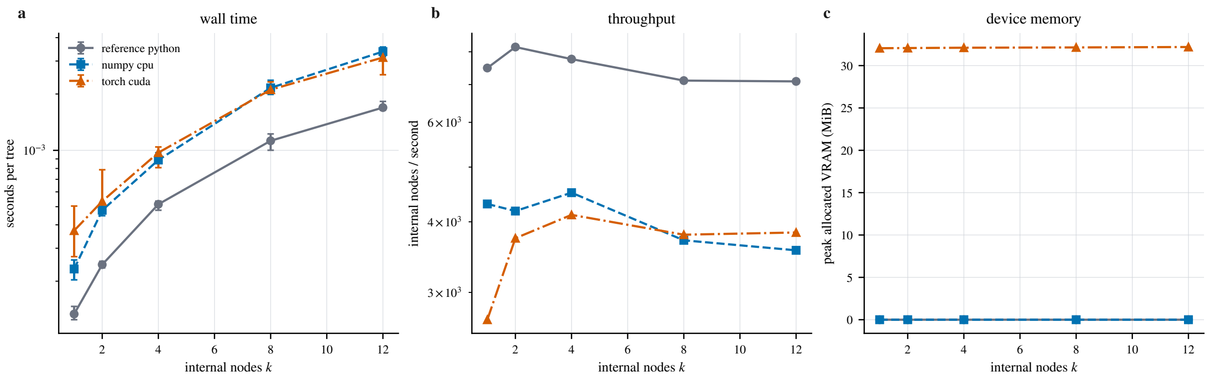

FIGURE 18. Wall-time benchmark on the hardware described in Appendix B: 15 backend and problem-size configurations, five timing repetitions each, 15–30 evaluations per repetition. Panels show median wall time with the observed minimum–maximum range, derived vertex throughput, and peak allocated device memory, for the reference coordinate evaluator, the vectorised CPU evaluator and the CUDA evaluator. Double-precision outputs agree exactly across the three. This is a measurement on one machine: it supports no asymptotic-complexity statement and no comparison between hardware platforms.

TABLE 18. Reproducibility manifest.

field	generated value
source commit	b718f4e51785
distinct configurations	15493
enumerated tree occurrences	81445
unique tree hashes	80870
closure-residual subsets	1530
CPU/GPU max difference	1.922e-08
optimizer trajectories requested	460800
optimizer status	not executed, under a stated computational-cost constraint
release status	FAIL_CLOSED_NOVELTY

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